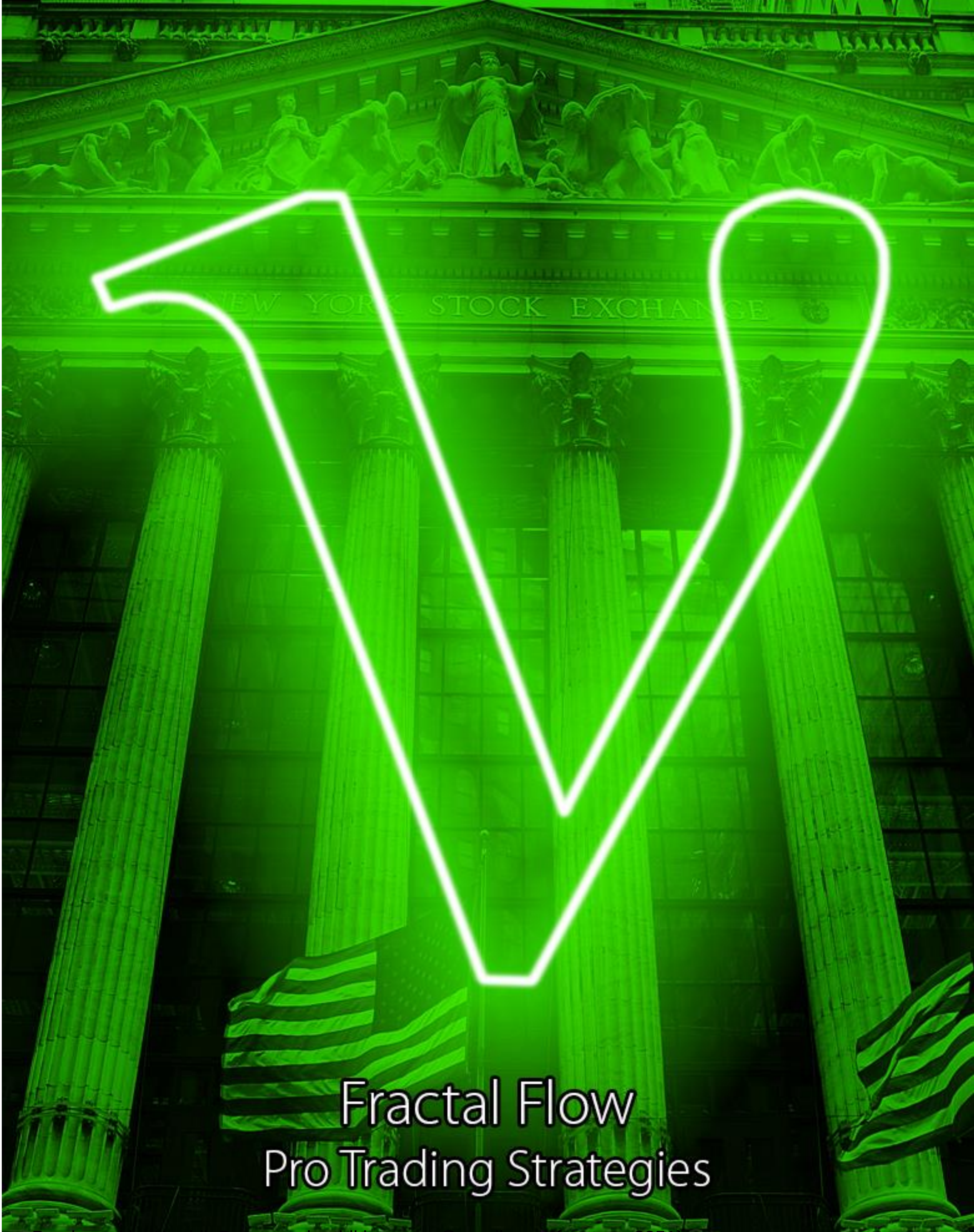

Fractal Flow
Pro Trading Strategies

Summary

Introduction

- What is Institutional Trading?

Institutional trading is the kind of trading done by funds, hedge funds, banks, and large companies. Each of these institutions vary in their trading goals, and also in the approach they use in the financial markets. It's surprising for the retail trader to learn that some institutions have different goals in the financial markets because most people believe that the only goal is to make money. However, in the institutional domain, things are more complicated than that.

For instance, the goal of a bank in the financial markets is not necessarily to make money, but to *transfer the risk* of the other operations that the bank engages in. The primary activity of a bank is to lend money, so they earn money with interest. These credit operations have a risk, which is the possibility of the borrower not being able to pay the loan plus interest. It's not difficult to imagine that the amount of risk that a large bank assumes is enormous. The bottom line is that such banks can use the financial markets to transfer some of this credit risk to other investors. Notice that this has nothing to do with making money per se. It's about optimally managing the risk of the credit operations the bank does.

Another example is of a large multinational company. These companies have the risk of currency devaluation when they firm contracts in the future because exchange rates fluctuate with some degree of unpredictability. This can work for as well as against the multinational company. One way of solving that problem is to use the derivative markets to hedge this risk or at least mitigate part of it, so the multinational company can engage in the purchase of currency options in the financial markets in order to hedge the risk of currency devaluation.

In the example of banks and large multinational companies, we are talking about using the markets as a ***hedging*** mechanism. The interesting thing is that, despite the fact that these institutions use hedging to protect themselves in different ways, ***the retail trader can use hedging to make money***. How that happens exactly is something that is only going to become clear as you apprehend the various institutional trading methods.

In the example of funds, the goal is indeed to make money to clients by generating an alpha over a benchmark. The point here is that the fund's clients don't need the fund if it doesn't offer any advantage over a market index. For example, if a fund is not going to yield a higher return than the S&P500, the client is simply better off buying the index directly. That's why funds attempt to generate alpha over a benchmark. In other words, the benchmark can be the S&P500, and the ***alpha*** is how much more than the benchmark the fund will return to its clients. Fund managers are paid proportionally to how large the alpha is, so they have all the incentive in the world to do a good job.

Funds usually employ a wide array of strategies using several different methods of trading in order to optimally diversify their risk. This is an important notion. When people talk about diversification, they think in terms of financial instrument, but that's a partial way of looking at it. Diversification means diversifying instruments, strategies, approaches, and the time horizon of strategies. A lot of what a fund does is related to speculation and arbitrage, which is conjecturing about the future direction of a financial instrument and exploiting the difference between markets respectively.

- Types of Institutional Trading

As it was already stated, there are mainly three types of institutional trading:

1. Speculation
2. Arbitrage
3. Hedging

Speculation is all about conjecturing about the future direction of a market. This is very familiar to retail traders because that's their general approach. A retail trader understands that to make money being long, the market has to rise, and to make money being short, the market has to fall. This might seem too obvious in a first moment, but later in the book you'll see this is not the only way of making money in the financial markets. There are situations where money can be made *independently* of what direction the market will go next, which is an insanely powerful idea.

Among the most common forms of speculation are fundamental analysis, technical analysis, macroeconomic analysis, application of certain econometric models that deal with timeseries forecasting, statistical models based on pattern recognition using machine learning and so on. The underlying principle behind these methods is the fact that certain economic forces or market patterns allow for a certain degree of predictability in the markets.

Arbitrage is a completely different approach. The idea of arbitrage is the exploitation of the *difference* between two or more markets, and this has little to do with the direction that these markets will go. The intuition of pure arbitrage is simple. If there are two markets that sell the same good at different prices, the *arbitrageur* can make money by buying the good in the market with lower price and selling it in the market with higher price. In today's electronic markets, this type of arbitrage is no longer practical due to the speed which information travels, but there are other types of arbitrage.

One of the most common types of arbitrage employed by funds, and that is also suitable for the retail trader, is called ***statistical arbitrage***. Roughly speaking, statistical arbitrage is exploiting the difference between two markets using a statistical model that describes one market in terms of the other using regression analysis. This will be the main topic of the book regarding institutional trading with arbitrage techniques. There are other types of arbitrage that hedge funds can do, but they are not suitable for the retail trader, so we are not going to waste time on those.

Hedging is the third type of institutional trading. As we already saw earlier, institutions use hedging in order to transfer risks from the institutions to investors/traders using the financial markets. The great epiphany here is that *retail traders can use the same hedging techniques to profit in the markets because they don't have to manage certain risks that only an institution has*. We have to be careful with this idea of hedging because it's widely misunderstood by retail traders. Hedging is done using the spot and derivatives markets in alignment and in specific ways. It's *not* about buying and selling *the same* financial instrument simultaneously, which just creates a redundant position.

Hedging involves the understanding of how derivatives work, and the intellectual *tour de force* that gave rise to the elegant mathematical models that allow several types of risks to be hedged or transferred in the financial markets. Hedging also allows the trader to minimize risks while maximizing returns in the most effective and scientifically sound way imaginable.

Trades involving the derivatives markets are considered to be one of the most sophisticated in all finance. Hopefully, you'll understand why that's the case with this series of books about institutional trading.

- Differences Between Institutional and Retail Trading

There are many differences between institutional and retail trading, and retail traders can benefit a lot by knowing what these differences are. The first is ***human capital***, which is the amount of knowledge present in institutions versus the knowledge that a single retail trader has. For instance, in a hedge fund, there are hundreds of employees that can research about the market, and teams that can make the best decisions. Each member can bring their own expertise not to mention that inside a hedge fund, the employees will have access to all kinds of techniques and operational information. The maximum a retail trader can do is to learn what science has to say about how to make money in the markets.

That's not common though. Most retail traders are after a simple fix or a simplistic formula that will allow them to make money consistently in the markets. Needless to say, this is a vain attempt. It's possible for a retail trader to apply some of the institutional methods, but it's absolutely necessary that this trader understands *at least* the practical side of what science has to say about the markets. Any attempt of finding a quick recipe is just a silly and immature attempt to cut corners.

Jim Simons - *the most successful hedge fund manager of all time* - only hires scientists for his firm Renaissance Technologies. It's not difficult to imagine the size of the intellectual gap between a hedge fund like that and a common retail trader who refuses to even accept that science has major contributions to the study of the financial markets. The tragedy is not that retail traders don't know the relevant scientific literature about finance. It's that they don't even know there is one in the first place.

Another big difference between institutional and retail traders is the ***infrastructure*** that is available to each one of them. Some institutional techniques can be applied by the retail trader, but others are simply impossible because they require a type of infrastructure that a retail trader will never have. For example, in quantitative funds, a common strategy is to swift through an enormous amount of data in order to find interdependencies that allow for short term prediction of the markets. That's incredibly expensive in terms of computational power. Only an institution is able to handle that level of data processing and analysis, not to mention the knowledge of how to actually do it, which is not simple either. Another curious example is the use of satellite data in order to gain some sort of edge in commodities trading. A retail trader can only dream about having access to this type of information.

Another big difference between retail and institutional trading is the amount of ***regulation*** involved in each one of them. Markets are not perfect, meaning that if there is no regulation, some companies eventually dominate everything or begin to implement unfair practices that might hurt other less competitive firms. The institutional trading domain is not an exception. Since institutions vary a lot in their capital power, the lack of regulation might create an unfair environment.

To give a practical example, there are certain regulations in certain countries that prohibit a fund to hold positions beyond a certain size, or to not have government bonds in the portfolio. These kinds of rules dramatically diminish the returns of a fund. Another potential problem is the fact that sometimes a fund has to close some of its positions in order to allow clients to withdraw their money. That can also affect the fund's performance negatively. *A retail trader has none of these problems*. He can trade his money in the way that he wishes.

Capital power is another great difference between retail and institutional traders. Institutions like funds and banks have a tremendous amount of capital at their disposal, and that means that a large sum of money can be generated with a small return. A retail trader that has a small amount of money will often employ leverage in order to magnify the returns, which is a *dangerous idea* for a variety of reasons.

- Challenges of Learning (and Teaching) Institutional Trading Techniques

At some point you might wonder why the institutional trading techniques and methods are not widely known if they are far more superior than the methods commonly employed by retail traders. The reason for that is that institutional trading methods have a lot of prerequisites, which function as a gate for people to go after them. In order to truly understand *the origin* of these methods, a person needs a good background in linear algebra, differential calculus, integral calculus, multivariable calculus, stochastic calculus, inferential statistics, microeconomics, macroeconomics, econometrics, and econophysics.

That statement alone is enough to scare most people away from these methods because most people want a simple solution to trade or invest their money. That's also why technical analysis became so popular among people who don't have a background in any of these prerequisites. In fact, there are almost no prerequisites to learn technical analysis, so it's obvious that most people gravitate towards it in a first moment. On the bright side, you don't need to be a master in all these prerequisites. *In fact, the reality is that you only need certain isolated ideas from each one in order to understand* ***the practice*** *of institutional trading*, which is what matters if you want to make money with them.

In other words, learning how to trade with institutional trading techniques is one thing. Learning how the complex mathematical and statistical models came to be is another completely. Another way of thinking about this is that it's possible to learn the intuition behind the complex formulas *without* having to learn how to solve them necessarily. That's a huge relief to people who want to learn these methods because they can learn the practice without having a PhD in math or statistics.

That requires good instruction though. This leads to my side of the challenge, which is to show how these methods work in the *simplest way possible* without overwhelming formulas or complex mathematical ideas. Only the rough and necessary intuition behind them. You'll see that in many cases I will show you a complicated formula, but I will tell you what the formula *means* for practical purposes. I believe this to be the sweet spot between learning and teaching these methods in the most effective way possible.

If you're one of those people who will quickly scan the book looking for "*the technique"*, keep in mind that I feel the need to show the mathematical models, and then later show what they mean. That will give you a sense that these formulas and concepts are not so complicated as they seem to be in mathematical notation. In fact, notation is one big barrier in this regard. When people don't understand the mathematical notation that is being used, a lot of confusion arises, and there is no way the person will be able to continue reading the material.

That's one of the problems that I will attempt solving in this book. I will show you the complex formulas, and then I will show you, *in simple terms*, what the formula means in practice. Here it's useful to remember the famous quote from Albert Einstein: *"Everything should be made as simple as possible, but not simpler"*. In other words, we should strive for simplicity, but we can't sacrifice quality for the sake of simplicity either. Another way of looking at this is that there is a *limit* to how much complex subjects can be simplified.

- The Retail Trader Advantage

We saw that there are many differences between retail and institutional traders. In some situations, retail traders can have an advantage though. If a retail trader learns the institutional methods, and has a decent capital to apply the methods, he will be in a better position than an institutional trader because he will have *the same level of knowledge while having no excess of regulations to deal with*. Beyond that, the retail traders can have the freedom (and responsibility) of trading their own money while an institutional trader would receive a salary and bonuses.

- Qualitative and Quantitative Analysis

In any institutional trading methodology, you'll find a duality. In other words, there is always a qualitative side and a quantitative side. The qualitative side is done by apprehending the economic context of what the market is doing or will do. The economic understanding will tell you if the market is going to rise or fall for example, but it doesn't say when, where or how much. The quantitative side of the analysis enters to solve exactly that problem. *It's the alignment between these two modes of analysis that produces good results*.

For example, in statistical arbitrage you'll learn the model for cointegration, but you cannot apply it everywhere expecting it will work just because it has a statistical edge. You still need to put the cointegration model within the current economic context. The same happens with the institutional methods that deal with the derivatives markets, even though some strategies have this qualitative aspect built into them. That's why I also wrote a book about economics for traders.

Some technical traders assume that economics has nothing to do with trading. That's a huge mistake of course because a good understanding of how the economic forces work can be the difference between making or losing money. Once again, traders who think like that are simply suffering from a strong temptation of keeping things simpler than they can be.

- The Ultimate Goal of Finance

The ultimate goal of finance is not so much about the goal itself, but *how* one achieves such goal, which is to make money of course. In other words, it's not just about making money, but ***making money in a way that is sustainable and safe in the long term***. The ultimate goal of finance is to minimize the variance of an equity curve, meaning minimizing risk as much as possible at the same pace that returns are maximized. Obtaining the *maximum return per unit of risk,* which is *very* different than obtaining high returns.

To understand that, let's imagine a simple example. Choose the alternative that you prefer:

A) 7% return with 5% of risk
B) 4% return with 2% of risk

Alternative A gives you 1.4% return per unit of risk while alternative B gives you 2% return per unit of risk. Even though alternative B yields a lower total return, it's a better option because it maximizes the return per unit of risk. This is a subtle example however, so it might not seem like B is indeed the best alternative, but that only serves to show how *our intuition is not tuned to make rational economic decisions*. The maximization of return per unit of risk is what allows the ***maximum possible return in the long term in the safest way possible***. Notice how that's very different than the maximum possible return in the short term in the most unsafe way possible. Again, it's this preoccupation with the time horizon of the economic decisions and their safety that leads to this idea of maximum return per unit of risk.

Many retail traders lose money on a consistent basis precisely because they don't understand how the time horizon of their decisions impacts their performance evaluation. Traders are sometimes able to get away with high risk in the short-term because the theoretical probability of their decisions hasn't had the time to converge to the experimental probability yet. In statistics, this is known as the *law of large numbers*. As time passes by, the theoretical probability of success converges with the experimental probability, so the trader who plays the high-risk approach *eventually* gets what he deserves.

Institutional methods use mathematics and statistics to enhance the theoretical probability of success, and also to guarantee that the trader will squeeze the maximum amount of return per unit of risk. That's the only way of trading consistently in the long term without unpleasant surprises in your equity curve.

- Validity of Institutional Trading Methods

At some point you might ask about the validity of the methods you are going to learn, and if you should *believe* them or not. The good news is that such methods are part of the scientific consensus about how the markets behave, so it's not a matter of believing them or not. They are the best that the scientific literature has to offer. I'm just facilitating your learning curve by creating a bridge between institutional and retail traders in a way that is accessible to most people. Many books that teach the institutional trading techniques are written with a degree of sophistication that makes it difficult to learn, and some authors seem to strive to being misunderstood in the effort of making their egos happier.

If you're not scientifically inclined, you may question the validity of science itself, and that's fair. It's important to realize that science is about eliminating human bias in a way that allows us to understand how the world works, which is not necessarily the same as how we perceive it to be or how we think it should be. The great German philosopher Immanuel Kant once pointed out that we cannot extract out of sense data the mechanism to make sense out of sense data. In other words, we perceive the world through a *filter*, which is our cognitive structure.

Our cognitive structure is the result of many variables like biological evolution, environment, personality, education, and so on. The bottom line is that different people perceive the world in different ways. Science is a way of eliminating this problem of relative perception in order to understand the world as it is, and not how it is perceived to be. The mechanism used for that is called the ***scientific method***. In very broad terms, science attempts to prove that an idea is *wrong* from many different perspectives. If this idea still survives this test, it has a good probability of being true.

Notice that this is quite different than trying to prove that something is right. When you attempt to prove that something is correct, you'll tend to focus on the part that supports your hypothesis, and ignore the part that goes against it. In other words, scientific progress is made by attempting to prove something is wrong and failing to do so. That also means that science is not perfect, but it's still *the best we have* to know how the world works since it's the most rigorous and impersonal method of understanding.

Economic decisions are filled with human bias because we definitely didn't evolve to make rational decisions when it comes to financial risk and time. This is why it's so important to incorporate science in the decision-making process. The lack of science creates a void that allows irrational and biased behavior to dominate, and when it comes to economic decisions, irrationality always loses in the end. Remember that for the most part of the biological evolution process, humans evolved to avoid lions in the savanna, not to trade in the markets, so our rough intuitions do a very poor job when it comes to trading.

Learning the institutional trading methods also comes with a certain peace of mind. That's because by using such methods, we are relying on the shoulders of giants. The concepts and ideas that form the backbone of the institutional trading methods are the end result of many smart people over hundreds of years culminating in a few Nobel Prizes that are surprisingly simple and elegant, but extremely powerful at the same time.

Basics of Derivatives

The knowledge of how derivatives work is paramount to the apprehension of several institutional trading methods and strategies. A ***derivative*** is a financial instrument that derives from another. For example, *futures and options*. For the practical purposes of this book, we will focus on options because they have many interesting properties related to risk. Options are a derivative because they are based on another financial instrument like stocks, currencies, commodities and so on. In this book we'll focus more on stock options for reasons that will become clear later on.

- Basics of Options

An option gives the trader *the right* to buy or sell the underlying asset (instrument on which the option is based) at a fixed price until or at a certain date called ***expiration date***. There are mainly two types of options. A ***call option*** gives the trader *the right to buy* the underlying asset at a fixed price until or at the expiration date, and a ***put option*** gives the trader *the right to sell* the underlying asset at a fixed price until or at the expiration date. For example, a Google Call Option gives the trader the right to buy the Google stock at a fixed price until or at the expiration date of that call option. In other words, when you buy a call option, you are buying the right to buy the underlying asset in the future at a fixed price.

This is why it's called an *option*. ***<u>A right to do something is not the obligation of doing something</u>***, meaning that the trader that buys an option has the right, but not the obligation to buy or sell the underlying asset at a fixed price in the future. This fixed price is known as the ***strike price*** of the option. If a trader chooses to buy or sell the underlying at the strike price, he chooses to exercise the option. With ***American Options***, the trader can exercise the option *at any moment* from now until the expiration date. With ***European Options***, the trader can only choose to exercise the option *at* the expiration date. These types of options have nothing to do with the geographical location where they are negotiated. They are just the names of such types of options. The ***option premium*** is how much the trader pays for the option, meaning how much he pays for the right, but not the obligation to buy or sell the underlying asset at a fixed price in the future.

It's important to point out that when options are traded, traders are negotiating *the right* to buy or sell the underlying asset. They are *not* negotiating the underlying asset itself. They wouldn't need options if they were going to negotiate the underlying asset. They could simply go to the spot market and negotiate the underlying asset directly.

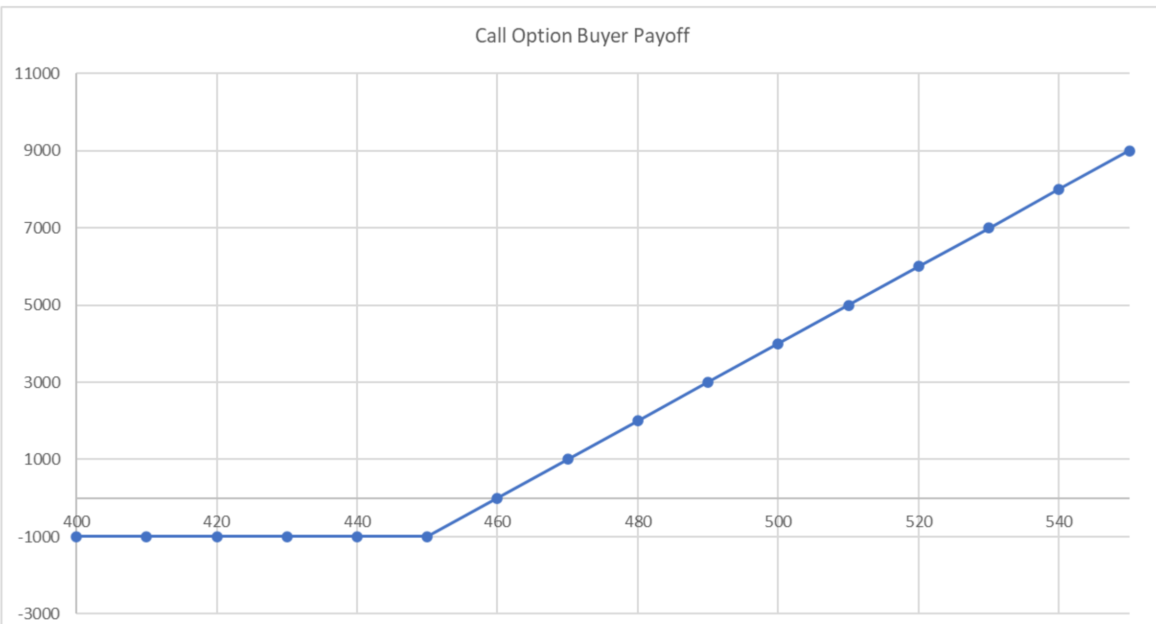
If a trader can buy the right to buy or sell the underlying asset at a fixed price in the future, it means that someone has to sell this right. That leads to the fact that there are four types of participants in the options market:

- Call Option Buyers
- Call Option Sellers
- Put Option Buyers
- Put Option Sellers

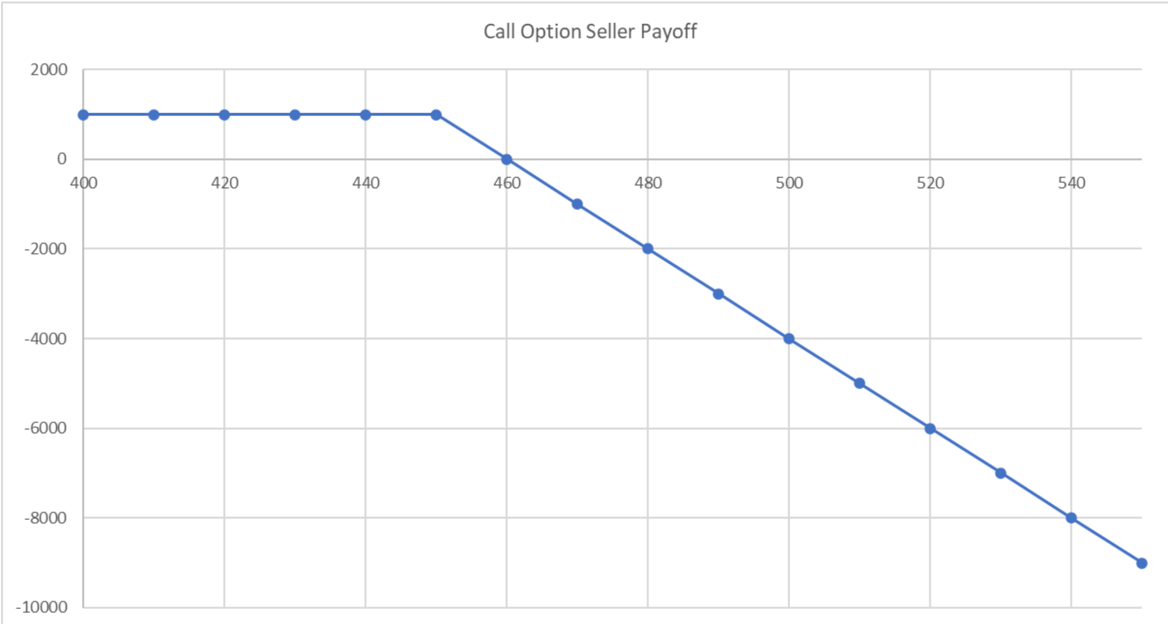
If a trader buys the right to buy the underlying (buys a call option), someone *must* sell the right to buy the underlying (sell a call option). Notice that there is an asymmetry between the buyers and sellers here. The *buyer has the right* to exercise the option, but the *seller has the obligation* to exercise this option if the buyer decides to do so. This is why short selling options is almost a sin in the institutional domain because it represents an infinite risk. Option buyers have *limited risk with unlimited potential reward* while option sellers have *unlimited risk with limited reward*.

The options market is not like the spot market where the relationship between buyers and sellers is symmetrical. This happens because, with options, one side of the market has a right, and the other has an obligation. The retail trader who is not used to derivatives might have difficulty understanding why this is the case, so it's useful to observe this asymmetrical relationship in graphical terms using payoff charts. Let's observe this intuition using call options, which are a little more intuitive than put options.

Let's assume that you buy 100 call options at $10 per option with the strike price of $450 to expire a month from now. That means you have the right to buy stock Y at $450 a month from now *despite* the actual price the stock will have at that time. If you decide to exercise your right, the trader in the other side of the transaction will have the obligation to sell the stock at $450. We can analyze these effects by observing the price of the stock at the expiration date. The horizontal axis represents the price at the expiration date. The vertical axis represents the profit:



Let's say that at the expiration date, the stock price is at $420. It doesn't make sense to exercise the call option with a strike of $450 because if you do it, you will have to buy the market at $450 and sell it at $420, which will generate a loss. The best case is to not exercise the option. The loss of $1000 at $420 is given by the option premium (100 x $10). If the price is at $450 at the expiration, it doesn't matter if you exercise the option or not because you will have a loss equal to the option premium you paid. If the stock price is above the strike price at the expiration, let's say at $490, it makes sense to exercise the option because then you will be able to buy the stock at $450 and sell it at $490 for a profit of $4000 minus the option premium of $1000. If you are in the buy side of the option, the graph makes it clear that your loss cannot be larger than $1000, but the profit can go up indefinitely. Observe now the call option seller payoff for the same transaction:



Like we saw previously, if the stock price is below the strike price for a call option, the call option buyer will not exercise it and lose the option premium. The option premium will be the profit for the call option seller. However, if the stock price is above the strike price at the expiration date, it's profitable for the call option buyer to exercise the option, so the call option seller will have to sell the stock at the strike price to the call option buyer despite the stock price at expiration. The higher the difference between the stock price and the strike price at expiration, the more the call option buyer profits, and the more the call option seller loses as we can see in the graph.

The bottom line is that buying options represent a limited risk with potentially infinite reward, and short selling options represent a potentially infinite risk with limited reward. You may ask why anyone would short sell options if that's the case. The problem is that some retail options traders don't understand this asymmetry between buying and selling options. Beyond that, there are certain short selling options strategies that allow the trader to profit from the simple passage of time. Even though this is true, the implicit risk of such strategies is infinite, so they are obviously a terrible idea in the long term. Note that the asymmetry is not between call and put options. It's between being on the buy side or sell side of call or put options.

It's necessary that you apprehend the basic structure of option to move on to the other more complex ideas. It's always useful to remember that when you buy options, you are *buying the right* to buy or sell the underlying. Making that distinction makes it easier to understand that you can buy the right to sell or sell the right to buy for example.

Stock Options Properties

There are many variables that affect the price of an option. In a first moment, we'll see how these variables work in a simple way, and then after we are going to see how these variables work together using an option pricing model. In the case of stock options, such model is the Black-Scholes model, which won the Nobel Prize in Economics in 1997. Even though the origins of the model and the journey that led to it has deep and complex roots, we'll try to apprehend its use in the simplest way *possible*. Let's begin by listing the variables that affect the price of an option first:

1. Underlying Stock Price (S_0)
2. Strike Price (K)
3. Time to Expiration (T)
4. Stock Price Volatility (σ)
5. Risk-Free Interest Rate (r)
6. Dividend Yield

- Underlying Stock Price and Option Strike Price

Let's begin by analyzing the relationship between the underlying stock price and the strike price of an option. In the case of a call option, the profit of exercising the option will be proportional to the quantity by which the stock price exceeds the strike price, so the higher the stock price is in relation to the strike price, the more valuable the call option becomes. For example, if the underlying stock price is at $40 at the expiration date, and a call option has a strike of $30, the trader can buy the stock at $30 by exercising the option, and then sell it at market by $40 with a profit of $10 minus the option premium.

If the underlying stock price is at $80, and the call option has the strike price of $30, it's obvious that the profit will be much larger, so the option will be more valuable. As the stock price increases in relation to the call option strike, the call option becomes more valuable. The higher the result from $S_0 – K$, the more valuable the call option.

In the case of the put option, the larger the result from $K – S_0$, the more valuable the put option. In other words, as the stock price decreases, a put option becomes more valuable. This leads us to the concept of intrinsic value of an option, and the different ways we classify the option based on the relationship between the underlying stock price and the option strike price.

The ***intrinsic value*** of an option is how much it is worth. For example, consider a put option with a strike price of $50 and an underlying stock price of $40 at the expiration date. The put option has an intrinsic value of $10 because at expiration, you can sell the market at $50 and buy it at $40 with a profit of $10 per share. When an option has intrinsic value, it's said that it is In-The-Money (ITM). A call option is ITM when the underlying stock price is above the strike price, and a put option is ITM when the underlying stock price is below the strike price.

An option is At-The-Money (ATM) when the underlying stock price and strike price are equal or at least very close to each other. A call option is considered to be Out-Of-The-Money (OTM) when the underlying stock price is below the strike price, where it doesn't make sense to exercise the option. A put option is considered to be OTM when the underlying stock price is above the strike price.

- Time to Expiration

Generally speaking, the larger the time to expiration, the more valuable an option is. That happens simply because if there is more time between the present and the expiration date, the higher the chances of the option becoming ITM.

- Volatility

Volatility is a measure of uncertainty about a market. The higher the volatility of a market, the higher the chances of price getting further away from the strike, and therefore, becoming deep in the money or deep out of the money. Whether or not volatility will be on the trader's side is a more complicated story, so we'll dedicate an entire section of the book to talk about that. Institutional trading with derivatives is trading the volatility of a stock, not the stock price.

- Risk-Free Interest Rate

As the risk-free interest rate increases, the expected returns of investors in a stock also increase because the quality of an investment is measured by the Sharpe Ratio, meaning the ratio between the expected return and the risk-free interest rate. Beyond that, the present value of the future cash flow in any stock decreases with an increase in the risk-free interest rate. The combined effect of these two situations make the value of a call option increase and a put option to decrease when there is an increase in the risk-free interest rate.

It's important to realize that we are observing the isolated effect of the risk-free interest rate on the value of options, meaning that the effects described above happen when all the other variables remain constant, which doesn't happen in real life. For example, when the risk-free interest rate decreases, the price of stocks tends to rise because people look for better investment alternatives than government bonds (usually in the stock market). In other words, if the risk-free interest rate falls, and the stock rises, that will positively affect call options and negatively affect put options.

However, the effect of the risk-free interest rates on options for the retail trader/investor is minimal, so there is no point worrying too much about that.

- Dividend Yield

The dividend yield has the effect of decreasing the price of stocks. If there are dividends between the present and the expiration date of an option, the value of options will be affected. Since dividends decrease the value of stocks, they will negatively affect the value of call options, and positively affect the value of put options.

The Black-Scholes Model

In the last chapter, we observed the effect that each variable has on the value of options. However, in the real world, all these variables change simultaneously, which makes it difficult to understand how the value of options evolve intuitively in real time. The Black-Scholes model is a mathematical formula that returns the fair price of a European option taking into account all of the variables we saw in the previous chapter. In a first moment, the retail investor might believe that the formula will tell if an option is overpriced or underpriced, but that's *not* how the formula works.

What actually determines the fair value of an option are the forces of supply and demand of the market, meaning the buyers and sellers of options. What happens is that the market participants assume that options can be correctly priced using the Black-Scholes model. That's very useful because it allows traders to understand the effect of the individual variables that affect the price of an option. We can observe that by obtaining the *partial derivatives* of the model called *option Greeks* (more on that later).

The story behind the Black-Scholes model goes back a few hundred years, when a botanist called Robert Brown observed the motion of pollen particles suspended in water (a normal experiment in botanic). Brown observed that the pollen particle displayed a random movement because it was constantly colliding with the water particles. This became known as ***Brownian Motion***. In the beginning of the twentieth century, a French mathematician called Louis Bachelier made a parallel between Brownian motion and the price of a financial asset. In his PhD thesis called "Theory of Speculation", he compared the movement of particles suspended in fluid (like the one Brown observed) with the movement of a financial asset, claiming that both were mathematically equivalent.

We can all understand why that's the case intuitively because just like the water particles collide constantly with the pollen particle in the botanic experiment, the many buyers and sellers collide with each other in a financial asset that is heavily traded. That's what creates the jittery motion of heavily traded stocks. That's the main intuition behind this parallel, and that's why it's assumed that the prices of stocks follow a Brownian Motion. A few decades later, a Japanese mathematician called Kiyoshi Ito described mathematically that this is in fact true in what became known as Ito's Lemma.

The Black-Scholes model is, roughly speaking, a natural consequence of the Ito's Lemma. In other words, it's a pricing model that assumes that stock prices indeed follow a Brownian Motion. The model also assumes that price returns are normally distributed, which is very useful. The genius of the model is precisely in this fact. The ability to model the random motion of prices that are assumed to be normally distributed. That is, among other reasons, why it won the Nobel Prize in 1997.

- How Traders Use the Black-Scholes Model

The Black-Scholes model has mainly two uses in the institutional domain, which is ***to extract the implied volatility of an option, and to calculate the partial derivatives of the model called option Greeks***. The implied volatility and the option Greeks are what allow traders to hedge different types of risk in different market scenarios. It's important to point out that you don't need to understand the deep origins of the Black-Scholes formula. You just need to understand the rough intuitions associated with it.

As it was stated previously, the Black-Scholes model returns the fair value of a European option, but since the fair value of the option is already determined by the supply and demand of the market, we can plug the value of the option into the formula to determine its implied volatility, which is nothing more than the volatility or risk implied in the stock from now until the expiration date of the option being traded. The historic volatility measures the standard deviation of log returns backwards in time. The implied volatility measures it *forwards* in time. All the other variables in the Black-Scholes formula are known. The model is described as follows:

$$\text{Call} = S_0 N(d_1) - Ke^{-rT} N(d_2)$$

And

$$\text{Put} = Ke^{-rT} N(-d_2) - S_0 N(-d_1)$$

Where:

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}}$$

Option Greeks

The option Greeks measure the different dimensions of risk when a trader assumes an options position. An *institutional trader* has the job of managing the Greeks in order to make the risk acceptable to the institution. Observe that the financial risk can be stripped down into other types of risks when we deal with options due to the multivariable nature of option premiums. The main job of an institutional trader in an options trading desk is to manage risk, not necessarily to create a profit. Retail traders however can make use of the same techniques to profit since they don't have the preoccupation of managing the risk of operations that arrive passively in their portfolios.

Option *Greeks* are extremely important to any trader who wishes to understand the behavior of options, and eventually make money with them. The option Greeks are the partial derivatives of the Black-Scholes model. The intuition behind partial derivatives is quite simple. When you have a multivariable equation like the Black-Scholes model, there are several variables affecting the result simultaneously. This makes it difficult to understand how the model functions as a whole.

Partial derivatives are ways of measuring how one variable changes given a minimal change in another variable with all else constant. In other words, a partial derivative tells how one variable in the model changes when another variable in the model changes, and all the remaining variables remain constant for simplicity sake. There are several partial derivatives we can take from the Black-Scholes model, but there are a few that are more important than others. The partial derivatives are called Greeks because each partial derivative receives a Greek letter associated with it.

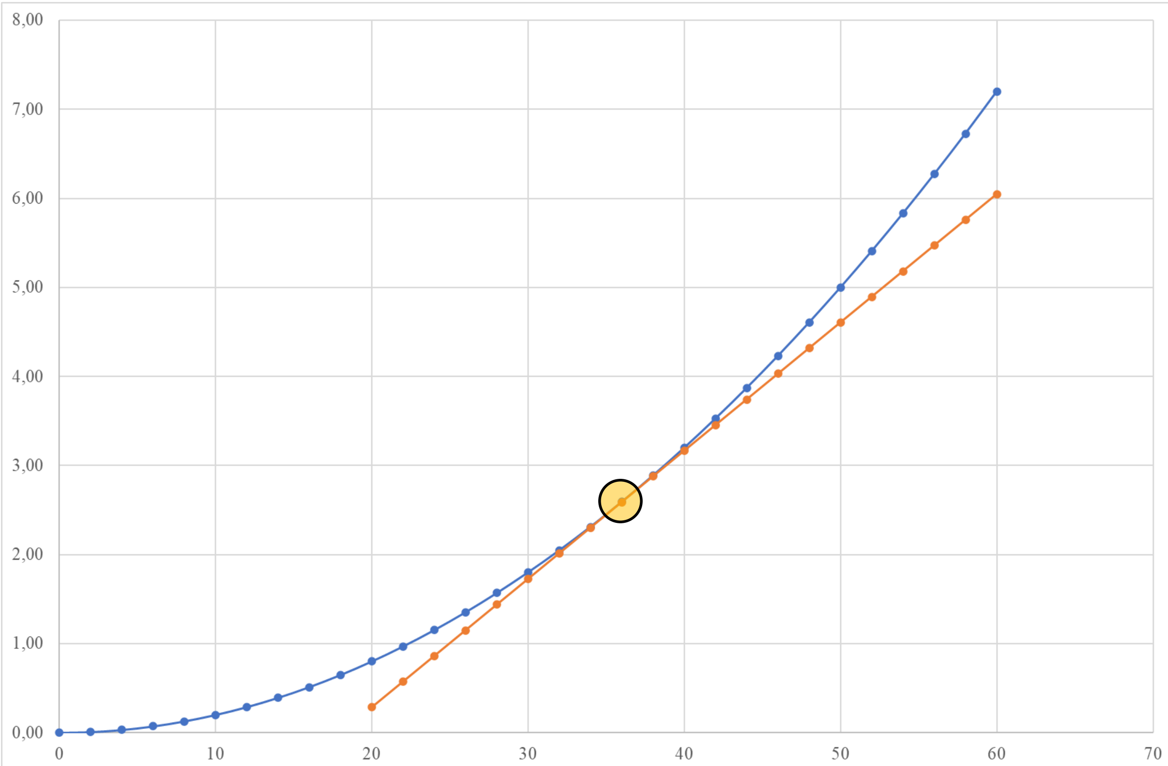
For example, the *Delta* of an option is, with everything else constant, *how much* the option premium changes in relation to changes in the underlying stock price. The Greeks we are going to worry about are the ***Delta, Gamma, Vega, and Theta***. These are the risk parameters that will allow you to eliminate or transfer certain risks in the market in very interesting and useful ways.

- Delta

The Delta is perhaps the most famous of option Greeks, and it measures the change in the option premium in relation to the change in the underlying stock price. Understanding the Delta is what allows the trader to *remove or hedge* the directional risk in the market. The formula for Delta of a call option is as follows:

$$\Delta = \frac{\partial c}{\partial S}$$

The symbol ∂ simply means *partial derivative*. The letter c relates to the call option premium (the same formula applies to the put option of course), and S to the underlying stock price. An important intuition about the Delta is that the underlying stock varies *linearly* while the option premium varies *nonlinearly*, so the Delta measures the inclination of the option premium curve as a function of the underlying stock price. We can understand this a little more intuitively by plotting the value of the option premium as a function of the underlying stock price:



Consider that the x-axis is the underlying stock price, and the y-axis is the option premium. Observe that as the stock price varies linearly, the option premium (blue curve) varies nonlinearly since it's a curve and not a straight line. The orange line measures the slope of the tangent line at x = 36. In other words, the Delta is the inclination of the option premium curve given an underlying stock price. This is important to understand how the delta allows the trader to remove the direction risk. Observe that at the underlying stock price of $36, the two lines meet. *Any value above $36 makes the difference between the blue and orange curves increase, but any value below $36 also does*. Another way of thinking about this is that the difference between the underlying stock price and the option premium increases despite the direction of the market.

More importantly, in terms of quantity, *buying X number of call options is the same as buying X number of shares of the underlying times the call option Delta*. That's the most important thing about the Delta of an option. In the same way, *buying X amount of put options is the same as selling X number of shares of the underlying times the delta of the put option*. It's important to realize that these things are equivalent in terms of quantity, but they are not equivalent in the way they vary. In terms of quantity, we have:

Buying X Call Options = Buying X Underlying Stock Shares *x* Call Option Delta

Buying X Put Options = Selling X Underlying Stock Shares *x* Put Option Delta

You might be wondering how exactly the Delta allows the trader to remove the directional risk of the market. The key to understand this is that even though you can buy or sell the same amount of stock using either the option or the stock, the way the option premium varies is different than the way the underlying stock varies. If we can use either the option or the stock to buy or sell the same amount of stock, we can create a hedge by buying options and selling the stock simultaneously. This is known as *Delta Hedge*.

In a Delta Hedge, the more the market rises, the *longer* your position becomes *without buying*. The more the market falls, the *shorter* your position becomes *without selling*, therefore removing or hedging the directional risk. This might seem like magic in a first moment, but it's just the difference in the way an option varies in relation to the underlying. In the institutional domain, derivatives are always traded using delta hedge. You can think of it as a starting position because there is no point being exposed to the directional risk of the market if you're going to use derivatives.

However, eliminating the directional risk now means that the financial asset being traded is *volatility, and not the stock price anymore*. That's because when a trader hedges the delta, he wants price to move as far away as possible from the price where he executed the hedge, so he is counting on the fact that the market will have a lot of volatility in the future. Observe that you are not worried about the direction that the market will take. You simply want the market to move *away* from the point where you hedged. *It doesn't matter if the market is going to go up or down. It just has to move*. <u>Notice that counting on the fact that the market is going to move is a lot easier than trying to predict which direction it will go.</u>

In summary, *a trader makes money with delta hedge when there is volatility*. It doesn't matter if price goes up or down as long as it moves away from the point where the delta hedge was made. The delta of an option is also the probability of the option being exercised, which implies the probability of the option becoming ITM. Another important idea is that the delta shows how powerful the other Greeks are. The closer the option delta is to 50%, the more pronounced the other Greeks. Later in the book you'll see mathematical proof that the delta eliminates the directional risk of the market, but for now let's stick with the intuition.

At this point you might be thinking that this is too good to be true, but we have to deal with other problems like the Vega and Theta. Before we talk about these Greeks, let's understand another Greek that relates to the Delta first: The Gamma.

- Gamma

The Gamma is simply the change of Delta in relation to the change in the underlying stock price. In other words, how much the delta varies with the movement in the underlying stock price:

$$\Gamma = \frac{\partial \Delta}{\partial S}$$

It's important to understand Gamma because of the *Gamma Effect*. When you buy options, you are entering a position called Positive Gamma, which basically means that any movement away from the hedge point (up or down) in the underlying is favorable to you. The higher the Gamma, the more a change in the underlying benefits you. When you short sell options, any movement in the underlying goes against you in what's called Negative Gamma.

This is why the Delta Hedge is done by buying options instead of short selling them. In basic terms, when you buy options, if there is chaos and volatility in the market, you win. When you short sell options, you win if the market remains calm without fluctuation. It's easy to see which one of these situations is more favorable. The market is always moving, and there is always some piece of news that agitates the market, so buying options is much better than short selling options in the context of a delta hedge.

Any time the market moves away from the point where the Delta was neutralized, the trader obtains a gain in Gamma (assuming he is in a Positive Gamma position). The higher the volatility in the market, the easier it will be to get gains in Gamma.

- Vega

Vega is the change in the option premium given a change in the implied volatility of the underlying stock:

$$\nu = \frac{\partial c}{\partial \sigma}$$

Delta Hedging with Positive Gamma is a really good idea, but we have to deal with certain problems that come along with that. One of these problems is called Vega, which is a Greek that measures the change in the option premium given a change in the implied volatility. We learned previously that volatility directly impacts the value of an option. The more volatility it has, the higher the chances of price moving away from the hedge point.

If the implied volatility decreases, the value of the option will also decrease. When the market remains quiet for a while without moving too much, the implied volatility will fall, and that will negatively affect the option value. Vega is not actually a problem. It's a favorable vector when there is volatility, and an unfavorable vector when there isn't volatility in the markets.

As we already established, a delta hedge is done by buying options because of the positive gamma effect. The consequence of that is a positive Vega, meaning that when the implied volatility rises, the P/L will be positively impacted, and when the implied volatility falls, the P/L will be negatively impacted. That makes sense because a positive Gamma implies that we want volatility to happen, so positive Gamma implies positive Vega. The Vega risk is one of the reasons that we need to buy volatility when it's low.

This elucidates the fact that a Delta Hedge is about trading volatility. If volatility rises, the trader will be able to obtain gains in Gamma, and the implied volatility will allow for gains in Vega as well. If volatility falls, the trader will have difficulty obtaining gains in Gamma, and he will obtain losses in Vega due to the fall in the implied volatility.

- Theta

Theta is simply the natural erosion in the option value as time passes by. As we saw earlier, the more time until expiration an option has, the higher its value because there is more implied volatility in it. The Theta measures the decay in the option premium given a change in the time until expiration. The Theta reflects the loss in implied volatility that happens with time:

$$\Theta = \frac{\partial c}{\partial t}$$

That means that options lose a little bit of their value just with the passage of time. This is one of the risks of Delta Hedging. *Your gains with Gamma and Vega have to surpass the losses with Theta*.

In a way, you are exchanging the directional risk of the market for the natural erosion of the option value (Theta), and for the possibility of not having volatility (Vega). The main point is that it’s much easier to deal with volatility than it is to deal with the direction of the market. As it was said earlier, we can heavily rely on the fact that there is going to be uncertainty, chaos and volatility in the financial markets. The question that remains is if the future volatility is going to be enough to produce gains in Gamma and Vega to surpass the losses in Theta.

- Buying Options vs Selling Options

We want to understand the difference between buying options and selling options from the point of view of Greeks. When we create a delta hedge by buying options, we have:

- Neutral Delta (the direction of the market doesn’t matter)
- Positive Gamma (any market movement away from the hedge is favorable to the P/L)
- Positive Vega (a rise in implied volatility positively impacts the P/L)
- Negative Theta (options lose value with the passage of time)

In other words, when an institutional trader is on the buy-side of a delta hedge, the direction the market will go doesn’t matter. Any market movement away from the point where the delta hedge was created positively impacts the P/L, and a rise in implied volatility also positively impacts the P/L. The simple passage of time erodes the value of options due to a loss in implied volatility. Another way of looking at this is that a delta hedge on the buy-side makes money when there is a lot of volatility in the markets because that means price will fluctuate to produce gains in Gamma and Vega that will be enough to surpass the losses in Theta. On the other hand, if we create a delta hedge by short selling options, we have:

- Neutral Delta (the direction of the market doesn’t matter)
- Negative Gamma (any market movement away from the hedge is unfavorable to the P/L)
- Negative Vega (a rise in implied volatility negatively impacts the P/L)
- Positive Theta (portfolio gains value with the passage of time)

Since it’s a delta hedge, the direction of the market doesn’t matter. However, a delta hedge on the sell-side creates a dangerous situation called negative gamma, meaning that any market movement away from the hedge negatively impacts the P/L, and any rise in volatility also negatively impacts the P/L. This is dangerous because the trader assumes that the market will not fluctuate and stay calm, which almost never happens. The sell-side delta hedge creates a positive Theta, meaning that the portfolio benefits from the simple passage of time. Some traders are attracted to this idea, but all we need to remember is that a positive Theta implies a negative Gamma, which is extremely dangerous. This is why a delta hedge is always done by buying options since traders want to benefit from the volatility in the markets, and not go against it.

- The Antifragility of Derivatives

In economics, there is the idea of fragility, which is the property of something being vulnerable to uncertainty, chaos, turmoil, volatility, and risk. Most people have an intuition that the opposite of fragility is robustness, but that's not the case. Robustness is the property of *surviving* chaos. The opposite of fragility is *antifragility*, which is the property of *thriving* with chaos. Notice that there is a big difference between surviving chaos and thriving with it. Options in the way that you are learning in this book are antifragile, meaning that they allow you to profit from the chaos of the market.

In any other type of trading strategy, chaos is something to fear. Trading strategies that don't involve derivatives and the hedging being proposing here are at best robust, meaning that they can survive the chaos in the markets, but they cannot thrive with it. That's an extremely powerful idea because, once again, problems are much more likely than solutions, destruction is much more likely than construction, and chaos is much more easily established than order. There is an asymmetry towards bad events, and the market reflects that all the time.

That's the idea of *entropy*, which is the natural tendency that things have to go out of order. In other words, effort is necessary just to maintain things as they are. Certainty and uncertainty dissipate in different ways. Certainty can easily vanish with one piece of bad news. Uncertainty on the other hand is much more stable and it takes longer to dissipate. Derivatives are a way of turning chaos (volatility) into a financial asset. They are basically a way of profiting from disorder.

- Trading Volatility

At this point, it should be clear that delta hedging is about trading volatility. That implies we want to enter the market when volatility is low with the expectation that it will increase. Delta hedging is about trading volatility because if we expect that volatility will rise, it means that we will probably benefit from the Gamma Effect and from gains in Vega. These two effects will surpass the losses with Theta if we buy volatility when it's low. That begs the question of how to know when volatility is indeed low.

Volatility is easier to deal with than price itself. That's because the direction of price can be highly misleading due to its Brownian nature. For example, before the release of an important macroeconomic indicator, it's difficult to predict the direction of price, but we can safely assume that volatility will increase. It's useful to make this dissociation between price direction and price volatility. There are different ways of knowing when volatility is cheap, but in the institutional domain, that's done by using an econometric model called GARCH(1,1). Volatility can also be anticipated in some situations like the earnings release of a stock, important macroeconomic indicators release, important events like elections and so on.

Delta

We already developed certain intuitions about how the partial derivatives work, but understanding how they behave in practice is paramount. To do that, we are going to use an excel sheet that allows us to simulate different market scenarios. The excel sheet we are going to use is from Macroption. We'll begin by developing certain practical intuitions about the Delta of an option. In these examples, I'm assuming that 1 option contract has 1 option, but that can change depending on what country you're in (1 contract usually has 100 options).

As it was stated in previous sections, the Delta is the change of the option price given a minimal change in the underlying stock price. The Delta is a number that varies from 0 to 1, so it can be expressed as a percentage. In practice, the Delta shows how much the option premium varies given a minimal change in the underlying stock price. The minimum a stock price can vary is 0.01. To understand that, consider the scenario below:

Option Strategy Simulator macroption

-1w	-1d	23/07/2020	+1d	+1w	-5%	-1%	Und Price	+1%	+5%	Div Yield	0,00%	Show Values & Greeks	-5%	-1%	Volatility	+1%	+5%
Now		13:59:15	Expiration		-1	-0.1	50,00	+0.1	+1	Int Rate	2,00%	Total	-1	-0.1	17,40%	+0.1	+1

Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price	Value	P/L	Delta	Gamma	Theta	Vega	Rho
1	1.000	Aug 2020	29,09	50	Call	1,01901	-1.019,01	1,01901	1.019,01	-0,00	522,73	162,21	-18,18	56,21	20,01
2		Jun 2020			None										
3		Jun 2020			None										
4		Jun 2020			None										
5		Jun 2020			None										
							-1.019,01		1.019,01	-0,00	522,73	162,21	-18,18	56,21	20,01

In this case, the underlying stock price is at \$50, and we are considering a call option with strike \$50. Notice that when the underlying stock price is at \$50, the option price is at \$1.01901 (we are going to consider several decimal cases to understand the effect of minimal changes in the option price). This particular call option has a Delta of 52.27%. This means that if the underlying price changes from \$50.00 to \$50.01, the option price should vary by 52.27% of \$0.01, which is \$0.00523, with all else maintained constant for simplicity sake:

Option Strategy Simulator macroption

-1w	-1d	23/07/2020	+1d	+1w	-5%	-1%	Und Price	+1%	+5%	Div Yield	0,00%	Show Values & Greeks	-5%	-1%	Volatility	+1%	+5%
Now		13:59:15	Expiration		-1	-0.1	50,01	+0.1	+1	Int Rate	2,00%	Total	-1	-0.1	17,40%	+0.1	+1

Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price	Value	P/L	Delta	Gamma	Theta	Vega	Rho
1	1.000	Aug 2020	29,09	50	Call	1,01901	-1.019,01	1,02424	1.024,24	5,23	524,35	162,14	-18,19	56,20	20,07
2		Jun 2020			None										
3		Jun 2020			None										
4		Jun 2020			None										
5		Jun 2020			None										
							-1.019,01		1.024,24	5,23	524,35	162,14	-18,19	56,20	20,07

Now that the underlying price is at \$50.01, the option price is at \$1.02424, which represents an increase of \$0.00523. Notice that, in practice, prices only have two decimal places, so sometimes the calculations are not entirely accurate because the level of accuracy decreases with a lower number of decimal places. The other important intuition about the Delta is that it is used to calculate the quantity of stock shares that correspond to a certain quantity of options.

In terms of quantity, buying x number of call options is equal to buying x number of stock shares times the delta of the call option at that moment. The Delta measures how long or how short the total position is. For example:

Option Strategy Simulator macroption

-1w	-1d	23/07/2020	+1d	+1w	-5%	-1%	Und Price	+1%	+5%	Div Yield	0,00%	Show Values & Greeks	-5%	-1%	Volatility	+1%	+5%
Now		13:59:15	Expiration		-1	-0.1	50,00	+0.1	+1	Int Rate	2,00%	Total	-1	-0.1	17,40%	+0.1	+1

Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price	Value	P/L	Delta	Gamma	Theta	Vega	Rho
1	1.000	Aug 2020	29,09	50	Call	1,01901	-1.019,01	1,01901	1.019,01	-0,00	522,73	162,21	-18,18	56,21	20,01
2		Jun 2020			None										
3		Jun 2020			None										
4		Jun 2020			None										
5		Jun 2020			None										
							-1.019,01		1.019,01	-0,00	522,73	162,21	-18,18	56,21	20,01

In the case above, we have 1000 call options in the portfolio. Notice that we have a positive portfolio Delta of 522.73. That means we are exposed to the direction of the market since we are delta positive. Since we bought call options and the portfolio is long, we need to sell a certain amount of stock shares in order to neutralize the portfolio delta. To calculate the amount of stock shares, we simply multiply the number of options (1000) by the delta of the option at that moment (52.27%), which is 1000 x 52.27 = 522.7. That will be rounded up to 523:

Option Strategy Simulator macroption

-1w	-1d	23/07/2020	+1d	+1w	-5%	-1%	Und Price	+1%	+5%	Div Yield	0,00%	Show Values & Greeks	-5%	-1%	Volatility	+1%	+5%
Now		13:59:15	Expiration		-1	-0.1	50,00	+0.1	+1	Int Rate	2,00%	Total	-1	-0.1	17,40%	+0.1	+1

Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price	Value	P/L	Delta	Gamma	Theta	Vega	Rho
1	1.000	Aug 2020	29,09	50	Call	1,01901	-1.019	1,01901	1.019	-0	522,73	162,21	-18,18	56,21	20,01
2	-523	Jun 2020			Stock	50,00	26.136	50,00	-26.136	0	-522,73				
3		Jun 2020			None										
4		Jun 2020			None										
5		Jun 2020			None										
							25.117		-25.117	-0	0,00	162,21	-18,18	56,21	20,01

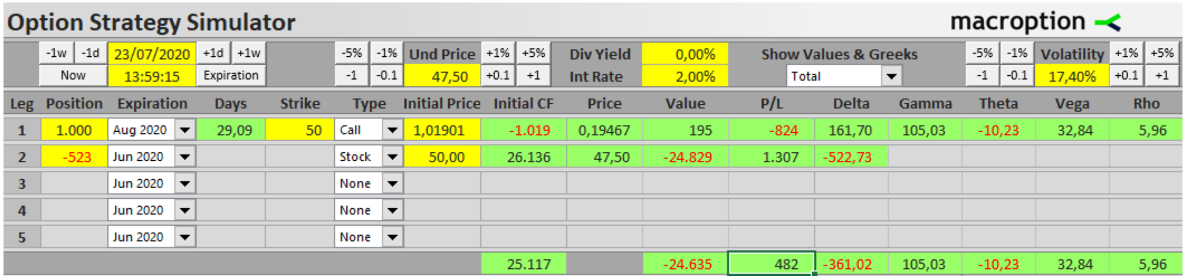
We bought 1000 call options and sold 523 shares of the underlying simultaneously. The Delta of our portfolio is now zero, meaning that we have a position that is *immune to the directional risk of the market*. To prove that's the case, let's observe what happens to the P/L if price goes up and if price goes down from this point. Let's first observe what happens if the underlying price goes up 5% from $50 to $52.50:

Option Strategy Simulator macroption

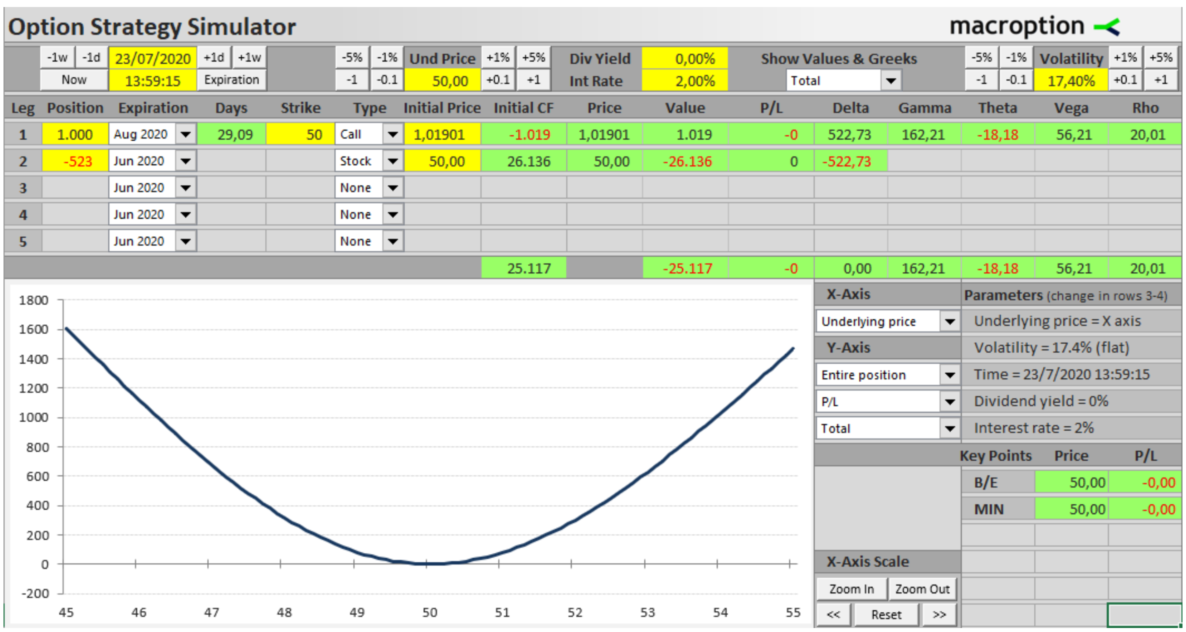
-1w	-1d	23/07/2020	+1d	+1w	-5%	-1%	Und Price	+1%	+5%	Div Yield	0,00%	Show Values & Greeks	-5%	-1%	Volatility	+1%	+5%
Now		13:59:15	Expiration		-1	-0.1	52,50	+0.1	+1	Int Rate	2,00%	Total	-1	-0.1	17,40%	+0.1	+1

Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price	Value	P/L	Delta	Gamma	Theta	Vega	Rho
1	1.000	Aug 2020	29,09	50	Call	1,01901	-1.019	2,77886	2.779	1.760	853,26	89,12	-12,48	34,04	33,47
2	-523	Jun 2020			Stock	50,00	26.136	52,50	-27.443	-1.307	-522,73				
3		Jun 2020			None										
4		Jun 2020			None										
5		Jun 2020			None										
							25.117		-24.664	453	330,53	89,12	-12,48	34,04	33,47

When price goes up 5%, that generates a positive P/L of $453. As a parenthesis, notice that when price moves up from the hedge point of $50, the delta becomes positive again (330.53), which means that our position is not neutral anymore. It's now a long position. In other words, when you hedge the delta of a portfolio and price goes up, the position becomes long as price rises *without the need of buying anything else*. Let's observe what happens now if price had fallen 5% from the original point of $50:



With price 5% down from $50 at $47.50, the P/L of our portfolio is *also positive* $482. Notice that now the total portfolio delta is negative (-361.02), meaning that as price moves down from the hedge point, the portfolio becomes short *without the need of selling anything else*. We can understand this more intuitively by looking at the payoff chart of a delta neutral position:



In the x-axis we have the underlying stock price, and in the y-axis, we have the P/L. Notice that the curve touches the x-axis at $50, which is the point where we neutralized the portfolio delta. If price rises from $50, the P/L increases. If price falls from $50, the P/L *also increases*. Observe also that linear changes in the x-axis produce nonlinear changes in the P/L. This is why a delta hedged position profits despite the direction of the market. Even though we can buy and sell the same quantity of stock with either options or the stock itself, the stock varies linearly while the options vary nonlinearly.

Notice that we are just observing the effect of one variable with all else being equal. In reality, several variables change simultaneously affecting the portfolio P/L in different ways. Even though the mathematics of a delta neutral position is perfect, there are other risks and factors we must consider while managing such portfolios. The understanding of these other risks will come with the other partial derivatives of the Black-Scholes model. The time until expiration, the implied volatility, the distance between the underlying price and the strike price, and the underlying stock price itself will affect the P/L in different ways.

In the institutional domain, any operation involving derivatives starts with a delta neutral position. That's because hedging the directional risk is simple, and there is no point being exposed to the directional risk of the market. In other words, eliminating the directional risk of the market is a simple thing to do, and it eliminates the problem of knowing if the stock is going to rise or fall, which turns out to be a difficult thing to do with a good amount of accuracy and timing in the long term. Another way of looking at this is that the delta of an option is a neutral vector on the portfolio. The other partial derivatives of a portfolio have to be understood assuming the position is delta neutral first.

Another important detail is that when an institutional trader creates a delta neutral position, he is not trading the stock price anymore. He is trading the stock volatility. That idea will become clearer as we observe how the other partial derivatives work.

Gamma

The gamma of an option measures the change in the option delta given a minimum change in the underlying stock price. We can also see gamma as the second derivative of the option price with respect to the underlying stock price. A positive gamma means that any change in the underlying stock price away from the hedge point is a positive vector on the portfolio P/L. A positive gamma also means that we are long on volatility, meaning that we want price to move as far as possible from the hedge point. Consider the same example from before:

Option Strategy Simulator macroption

-1w -1d	23/07/2020	+1d +1w	-5% -1%	Und Price	+1% +5%	Div Yield	0,00%	Show Values & Greeks	-5% -1%	Volatility	+1% +5%
Now	13:59:15	Expiration	-1 -0.1	50,00	+0.1 +1	Int Rate	2,00%	Total	-1 -0.1	17,40%	+0.1 +1

Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price	Value	P/L	Delta	Gamma	Theta	Vega	Rho
1	1.000	Aug 2020	29,09	50	Call	1,01901	-1.019	1,01901	1.019	-0	522,73	162,21	-18,18	56,21	20,01
2	-523	Jun 2020			Stock	50,00	26.136	50,00	-26.136	0	-522,73				
3		Jun 2020			None										
4		Jun 2020			None										
5		Jun 2020			None										
							25.117		-25.117	-0	0,00	162,21	-18,18	56,21	20,01

Notice that our portfolio is delta neutral, and it has a positive gamma (162.21). If we produce a minimal change in the underlying stock price from \$50.00 to \$50.01, the portfolio delta will increase by the gamma times 0.01:

Option Strategy Simulator macroption

-1w -1d	23/07/2020	+1d +1w	-5% -1%	Und Price	+1% +5%	Div Yield	0,00%	Show Values & Greeks	-5% -1%	Volatility	+1% +5%
Now	13:59:15	Expiration	-1 -0.1	50,01	+0.1 +1	Int Rate	2,00%	Total	-1 -0.1	17,40%	+0.1 +1

Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price	Value	P/L	Delta	Gamma	Theta	Vega	Rho
1	1.000	Aug 2020	29,09	50	Call	1,01901	-1.019	1,02424	1.024	5	524,35	162,14	-18,19	56,20	20,07
2	-523	Jun 2020			Stock	50,00	26.136	50,01	-26.141	-5	-522,73				
3		Jun 2020			None										
4		Jun 2020			None										
5		Jun 2020			None										
							25.117		-25.117	0	1,62	162,14	-18,19	56,20	20,07

When the underlying stock price was \$50, the gamma was 162.21. When price increased by 0.01, the portfolio delta increased by gamma times 0.01 (162.21 x 0.01 = 1.62). A positive gamma means that we gain from an increase in volatility, which will translate into price moving away from the hedge point. We can only create a positive gamma by buying options. An institutional trader wants the positive gamma to be as high as possible because that positively impacts the effect of price fluctuations on the P/L.

The positive gamma is the reason a delta neutral position is built by buying options instead of short selling them. As it was stated before, when a trader buys options, he is buying a right to negotiate the underlying stock in the future at a fixed price. That represents a limited risk and a potentially infinite reward. The trader who short sells an option creates a negative gamma position, meaning that he is short on volatility. That represents an infinite risk and limited reward. *In the institutional domain, short selling options is almost prohibited because of the negative gamma.*

In other words, when a trader has positive gamma, he is counting that the market will display volatility, chaos and uncertainty, meaning that he is long on volatility. However, when a trader has negative gamma, he is counting on the possibility that the market will not fluctuate, and that there is not going to be volatility or fluctuations. It's clear that this is a bad idea because *one of the certainties about the financial markets is that there is going to be uncertainty, volatility, chaos, and fluctuation*. The question is *how much* volatility will happen.

The duality between positive and negative gamma positions allows us to apprehend how a delta neutral position creates the possibility to profit from chaos in the markets, which is something unthinkable for purely speculative positions on the stock market. Without derivatives, a stock market position is at best robust in relation to chaos, meaning that it can survive it. A delta neutral position with positive gamma allows the trader to *gain* from chaos instead.

So far, we saw that delta is a neutral vector, and by buying options, we create a positive vector with gamma. However, a positive gamma implies that there is going to be a negative theta.

Theta

The Theta of an option is the change of the option price in relation to a change in the time until expiration. In other words, the Theta measure the natural erosion in the option price as time passes by. A position with positive gamma generates a negative theta, meaning that when we are long on volatility, the simple passage of time erodes the value of the option. That happens because the closer an option gets to expiration, the less implied volatility it has, and therefore, the less valuable it becomes. This represents a challenge because the positive vector created by gamma goes against the negative vector created by theta. That means that price has to move away from the hedge point enough to create gains in Gamma in order to surpass the losses in Theta. Consider the original hedged position at $50 once again:

Option Strategy Simulator macroption

-1w	-1d	23/07/2020	+1d	+1w	-5%	-1%	Und Price	+1%	+5%	Div Yield	0,00%	Show Values & Greeks	-5%	-1%	Volatility	+1%	+5%
Now		13:59:15	Expiration		-1	-0.1	50,00	+0.1	+1	Int Rate	2,00%	Total	-1	-0.1	17,40%	+0.1	+1

Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price	Value	P/L	Delta	Gamma	Theta	Vega	Rho
1	1.000	Aug 2020	29,09	50	Call	1,01901	-1.019	1,01901	1.019	-0	522,73	162,21	-18,18	56,21	20,01
2	-523	Jun 2020			Stock	50,00	26.136	50,00	-26.136	0	-522,73				
3		Jun 2020			None										
4		Jun 2020			None										
5		Jun 2020			None										
							25.117		-25.117	-0	0,00	162,21	-18,18	56,21	20,01

The portfolio has a negative theta (-18.18). That means that a day from now, the natural erosion of the option premium will negatively affect the P/L by -18.18. Observe what happens to the P/L from as we change the date from 23/07/2020 to 24/07/2020:

Option Strategy Simulator macroption

-1w	-1d	24/07/2020	+1d	+1w	-5%	-1%	Und Price	+1%	+5%	Div Yield	0,00%	Show Values & Greeks	-5%	-1%	Volatility	+1%	+5%
Now		13:59:15	Expiration		-1	-0.1	50,00	+0.1	+1	Int Rate	2,00%	Total	-1	-0.1	17,40%	+0.1	+1

Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price	Value	P/L	Delta	Gamma	Theta	Vega	Rho
1	1.000	Aug 2020	28,09	50	Call	1,01901	-1.019	1,00068	1.001	-18	522,33	165,08	-18,48	55,23	19,32
2	-523	Jun 2020			Stock	50,00	26.136	50,00	-26.136	0	-522,73				
3		Jun 2020			None										
4		Jun 2020			None										
5		Jun 2020			None										
							25.117		-25.136	-18	-0,39	165,08	-18,48	55,23	19,32

With all else equal, just from the passage of time, the P/L is now negative. This is why a delta neutral position has to move away from the hedged position. In other words, the gains in gamma have to surpass the losses in theta. Some retail option traders attempt to gain from the passage of time by short selling options. Even though short selling options will generate a positive theta, it will also generate a negative gamma, which is too dangerous.

The risk of a delta neutral position is price not moving away from the hedge point. If price doesn't move away, the theta will erode the option value with time, and the implied volatility will start to fall. To understand the effects of the implied volatility in the portfolio, we need to understand Vega.

Vega

The Vega measures the change in the option value given a change in the implied volatility of the option. The Vega can be a positive or a negative vector on the portfolio. If we have positive Gamma, the Vega will be positive, meaning that an increase in the implied volatility will positively affect the P/L:

Option Strategy Simulator macroption

-1w	-1d	23/07/2020	+1d	+1w	-5%	-1%	Und Price	+1%	+5%	Div Yield	0,00%	Show Values & Greeks	-5%	-1%	Volatility	+1%	+5%
Now		13:59:15	Expiration		-1	-0.1	50,00	+0.1	+1	Int Rate	2,00%	Total	-1	-0.1	17,40%	+0.1	+1

Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price	Value	P/L	Delta	Gamma	Theta	Vega	Rho
1	1.000	Aug 2020	29,09	50	Call	1,01901	-1.019	1,01901	1.019	-0	522,73	162,21	-18,18	56,21	20,01
2	-523	Jun 2020			Stock	50,00	26.136	50,00	-26.136	0	-522,73				
3		Jun 2020			None										
4		Jun 2020			None										
5		Jun 2020			None										
							25.117		-25.117	-0	0,00	162,21	-18,18	56,21	20,01

In the same example we have been using so far, the implied volatility remained constant at 17.40%. Notice also that the portfolio Vega is 56.21. That means that a 1% increase in the implied volatility will positively affect the P/L by 56.21:

Option Strategy Simulator macroption

-1w	-1d	23/07/2020	+1d	+1w	-5%	-1%	Und Price	+1%	+5%	Div Yield	0,00%	Show Values & Greeks	-5%	-1%	Volatility	+1%	+5%
Now		13:59:15	Expiration		-1	-0.1	50,00	+0.1	+1	Int Rate	2,00%	Total	-1	-0.1	18,40%	+0.1	+1

Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price	Value	P/L	Delta	Gamma	Theta	Vega	Rho
1	1.000	Aug 2020	29,09	50	Call	1,01901	-1.019	1,07522	1.075	56	522,59	153,40	-19,15	56,21	19,96
2	-523	Jun 2020			Stock	50,00	26.136	50,00	-26.136	0	-522,73				
3		Jun 2020			None										
4		Jun 2020			None										
5		Jun 2020			None										
							25.117		-25.061	56	-0,14	153,40	-19,15	56,21	19,96

Observe that when the implied volatility goes from 17.40% to 18.40%, the P/L increases by 56. If price is volatile, Vega will be a positive vector. If price is calm and doesn't fluctuate over time, Vega will be a problem. Once again, in the institutional domain, a delta neutral position is made on the buy-side of options. That generates the following scenario:

- A neutral Delta.
- A positive Gamma.
- A positive Vega.
- A negative Theta.

In summary, the Buy-Side Delta Hedge operation means that the direction the market will go is irrelevant; the fluctuations of the market away from the hedge point positively affect the P/L; an increase in implied volatility positively affects the P/L; and the passage of time negatively affects the P/L. In other words, we want to buy volatility when it's relatively low, and get out of the operation when it's relatively high. Before moving on to the different approaches to determine if volatility is cheap or expensive, let's observe another detail about the way we structure a delta hedge.

Greeks & Time

In the institutional domain, traders think about derivatives trades in terms of the Greeks, so whenever a particular market scenario occurs, the trader must be able to understand intuitively how to use the option Greeks to manage the situation in the best way possible. That leads us to the effect of Greeks in relation to the time until expiration of an option. The different option Greeks behave in different ways in relation to the time until expiration. In general terms, the closer an option is to the expiration date, the more powerful the Gamma and Theta, and the less powerful the Vega.

In other words, if you are trading an option that is close to the expiration date, the Gamma and Theta effects will be much more powerful than the Vega. That can be an interesting possibility when we want to minimize the effects of changes in implied volatility in our trades, and maximize the effects of Gamma. The more time until expiration an option has, the more implied volatility it has, so the Vega effect is more pronounced. On the other hand, the Gamma and Theta are less pronounced.

Delta Hedge without Stocks

So far, we saw how to neutralize the Delta using the underlying stock price, but there is another way of doing it. We can create a Delta Hedge by using derivatives only, meaning buying a specific combination of call and put options. To do that, we need to recall that buying calls is equal to buying the stock by the call delta, and buying puts is the same as selling the stock by the put delta. If we sum the deltas of call and put options of the same strike and same expiration, they will sum to 100%.

If we want to create a delta neutral position using only options, we must determine the total amount of options we want in the portfolio. The number of calls will be determined by the total number of options in the portfolio times the put delta, and the number of puts will be determined by the total number of options by the call delta. Let's observe an example:

Option Strategy Simulator macroption

-1w	-1d	23/07/2020	+1d	+1w	-5%	-1%	Und Price	+1%	+5%	Div Yield	0,00%	Show Values & Greeks	-5%	-1%	Volatility	+1%	+5%
Now		14:00:00	Expiration		-1	-0.1	50,00	+0.1	+1	Int Rate	2,00%	Total	-1	-0.1	17,40%	+0.1	+1

Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price	Value	P/L	Delta	Gamma	Theta	Vega	Rho
1	5.000	Aug 2020	29,09	50	Call	1,0190	-5.095,00	1,0190	5.095,00	-0,00	2.614	811	-91	281	100
2	5.000	Aug 2020	29,09	50	Put	0,9394	-4.697,00	0,9394	4.697,04	0,04	-2.386	811	-77	281	-99
3		Jun 2020			None										
4		Jun 2020			None										
5		Jun 2020			None										
							-9.792,00		9.792,04	0,04	227	1.622	-168	562	1

Let's say we want to have 10,000 options in the portfolio. In a first moment, the trader might think that to neutralize the delta, he has to buy an equal amount of calls and puts, but notice that the portfolio delta is positive (227). That happens because the call delta is 52.27%, and the put delta is 47.73% in this case. To neutralize the delta, we must buy 10,000 calls times the put delta, and 10,000 puts times the call delta:

Option Strategy Simulator macroption

-1w	-1d	23/07/2020	+1d	+1w	-5%	-1%	Und Price	+1%	+5%	Div Yield	0,00%	Show Values & Greeks	-5%	-1%	Volatility	+1%	+5%
Now		14:00:00	Expiration		-1	-0.1	50,00	+0.1	+1	Int Rate	2,00%	Total	-1	-0.1	17,40%	+0.1	+1

Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price	Value	P/L	Delta	Gamma	Theta	Vega	Rho
1	4.773	Aug 2020	29,09	50	Call	1,0190	-4.863,69	1,0190	4.863,68	-0,00	2.495	774	-87	268	95
2	5.227	Aug 2020	29,09	50	Put	0,9394	-4.910,24	0,9394	4.910,29	0,04	-2.495	848	-81	294	-103
3		Jun 2020			None										
4		Jun 2020			None										
5		Jun 2020			None										
							-9.773,93		9.773,97	0,04	0	1.622	-168	562	-8

The total amount of options in the portfolio is still 10,000 (4,773 + 5,227), but the equilibrium between calls and puts is tilted in such a way that neutralizes the delta. There are advantages and disadvantages in doing this. The advantage is that it is a lot cheaper to buy puts than it is to sell the underlying stock, meaning that the trader needs a lot less margin to perform the same operation. The other detail, which can be good or bad, is that when we neutralize the delta with options only, *we add the effect of Greeks to the portfolio*, meaning that they become more pronounced. That makes sense because before we only had the call Greeks to manage. Now we have call and put Greeks on top of each other.

Notice that a trader would need $9,773.97 to buy the calls and puts for this operation, which would generate a Gamma of 1,622; a Theta of -168; and a Vega of 562. Notice what happens if we decide to neutralize the delta in this operation using the underlying stock instead of the put options:

Option Strategy Simulator macroption

-1w	-1d	23/07/2020	+1d	+1w	-5%	-1%	Und Price	+1%	+5%	Div Yield	0,00%	Show Values & Greeks	-5%	-1%	Volatility	+1%	+5%
Now		14:00:00	Expiration		-1	-0.1	50,00	+0.1	+1	Int Rate	2,00%	Total	-1	-0.1	17,40%	+0.1	+1

Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price	Value	P/L	Delta	Gamma	Theta	Vega	Rho
1	4.773	Aug 2020	29,09	50	Call	1,0190	-4.864	1,0190	4.864	-0	2.495	774	-87	268	95
2	-2.495	Aug 2020			Stock	50,0000	124.748	50,0000	-124.748	0	-2.495				
3		Jun 2020			None										
4		Jun 2020			None										
5		Jun 2020			None										
							119.885		-119.885	-0	0	774	-87	268	95

We would need the margin to short sell 2,495 stock shares at $50 ($124,748) to neutralize the Delta of the same 4,773 call options, generating a smaller Gamma of 774; a Theta of -84, and a Vega of 268. In that sense, a Delta neutral position using only derivatives is naturally leveraged.

Gamma-Theta Breakeven

At this point it should be clear that the gains in Gamma and Vega should surpass the losses in Theta. The Vega is a special case because it can work for the trader as well as against him, so we'll deal with it later. However, when we hedge the delta on the buy side of options, the Gamma will always be positive, and the Theta will always be negative. *In the institutional domain, a good options trader is the trader that consistently pays for the Theta using the gains in Gamma*. Delta hedge operations are always D+0 or D+1, meaning that the operation is either opened and closed in the same day (D+0) or it is opened in one day, and closed in the next day (D+1). However, a D+1 operation can become D+0 if the trader manages to pay for the Theta of one day passing by.

To pay for Theta, the underlying stock price has to move away from the hedge point in order to generate a gain in Gamma. *The Gamma-Theta Breakeven is how much the underlying stock price has to vary in order to generate a gain in Gamma that will cover the costs of Theta from one day to the other*. If the trader pays for the Theta from one day to the other, he is able to carry the Delta hedge an extra day, therefore exposing himself to a possible increase in volatility (maybe even a Black Swan), which is exactly what he wants to happen.

Consider once again the scenario below where we create a delta neutral portfolio by buying call options and short selling the underlying stock price:

Option Strategy Simulator — macroption

-1w	-1d	23/07/2020	+1d	+1w	-5%	-1%	Und Price	+1%	+5%	Div Yield	0,00%	Show Values & Greeks	-5%	-1%	Volatility	+1%	+5%
Now		14:00:00	Expiration		-1	-0.1	50,00	+0.1	+1	Int Rate	2,00%	Total	-1	-0.1	17,40%	+0.1	+1

Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price	Value	P/L	Delta	Gamma	Theta	Vega	Rho
1	4.773	Aug 2020	29,09	50	Call	1,0190	-4.864	1,0190	4.864	-0	2.495	774	-87	268	95
2	-2.495	Aug 2020			Stock	50,0000	124.748	50,0000	-124.748	0	-2.495				
3		Aug 2020			None										
4		Jun 2020			None										
5		Jun 2020			None										
							119.885		-119.885	-0	0	774	-87	268	95

Here we can see that the Theta is equal to -87, meaning that a day from now, the P/L will suffer an erosion of -$87. To calculate the Gamma-Theta Breakeven, we must calculate how much the underlying stock price must vary in order to generate a gain in Gamma that will translate into $87 to neutralize the Theta. We can easily do this in Excel using the "Goal Seek" function. We define the P/L cell to 87 altering the underlying stock price cell:

Option Strategy Simulator — macroption

-1w	-1d	23/07/2020	+1d	+1w	-5%	-1%	Und Price	+1%	+5%	Div Yield	0,00%	Show Values & Greeks	-5%	-1%	Volatility	+1%	+5%
Now		14:00:00	Expiration		-1	-0.1	50,48	+0.1	+1	Int Rate	2,00%	Total	-1	-0.1	17,40%	+0.1	+1

Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price	Value	P/L	Delta	Gamma	Theta	Vega	Rho
1	4.773	Aug 2020	29,09	50	Call	1,0190	-4.864	1,2863	6.139	1.276	2.858	745	-86	263	110
2	-2.495	Aug 2020			Stock	50,0000	124.748	50,4764	-125.937	-1.189	-2.495				
3		Aug 2020			None										
4		Jun 2020			None										
5		Jun 2020			None										
							119.885		-119.798	87	363	745	-86	263	110

The image above shows that the Gamma-Theta Breakeven occurs when price reaches \$50.48, which is a 0.96% change. That means that if price moves away 0.96% *in either direction*, the Theta will be neutralized from one day to the other, and the trader will be able to carry the Delta Hedge portfolio to the next day. The breakeven point will vary according to the parameters, but it doesn't change that much from around 1% in the underlying stock variation.

The way to properly manage the Theta in a Delta neutral position is by neutralizing the portfolio Delta once the Gamma-Theta breakeven is reached or when half breakeven is reached. Sometimes it's easier to pay for half breakeven twice than it is to pay one breakeven at once.

Vega Risk

As we saw already, Vega can be a positive or negative vector on the portfolio. If the implied volatility rises, we will benefit from Vega. If the implied volatility falls, we will suffer a cost from Vega. *All of that implies that we must perform a delta hedge when the implied volatility is cheap*. That way we can benefit from Vega, and the occasional gains in Gamma that a rise in volatility will cause. One of the most common ways to forecast volatility in the institutional domain is by using an econometric model called GARCH(1,1). However, before understanding the GARCH(1,1) model, we need to have a good grasp on what volatility means in the first place.

Forecasting Volatility

The trader must be able to distinguish between different measures of volatility in order to know if volatility is cheap or expensive, which is the core foundation of a simple volatility trade. When the trader is able to buy cheap volatility, the gains in Vega and in Gamma will most likely surpass the losses in Theta, which is how a delta neutral position becomes profitable. Remember once again that this has nothing to do with the direction the market will go. The financial asset being traded is the *underlying stock volatility*, not the underlying stock price. Before moving on to the different ways of measuring volatility, let's understand what it means mathematically and statistically.

Volatility, risk, and standard deviation mean *the same thing* in finance. When a trader wonders about the risk of a financial asset or the volatility of a financial asset, he is wondering about the same question. There are mainly two types of volatility:

- Historical Volatility
- Implied Volatility

The ***implied volatility*** is the volatility that is implied in the underlying stock from now until the expiration of a given option, and it's obtained through the Black-Scholes model as it was already mentioned. You can think of the implied volatility as the future volatility (assuming that stock returns are normally distributed, and that price follows a Brownian motion). The ***historical volatility*** measures the past volatility of a stock, and it is measured by the *standard deviation of log returns of a stock*. We need to develop a few intuitions to understand that statement. More specifically, we need to understand the difference between the stock price and the stock returns; the difference between the simple return and the log return; and the intuition of standard deviation, and why it is equal to volatility and risk. It's the *comparison* between different measurements of historical and implied volatility that will allow the trader to determine if volatility is cheap or expensive like an institutional trader does.

- Stock Price vs Stock Return

The first basic intuition is of the simple return, which measures the percentage change from one period to the next. For example, if the stock price today is \$102, and it was \$100 the day before, we calculate the simple return as follows:

$$\text{Simple Return} = \frac{S_t - S_{t-1}}{S_{t-1}}$$

S_t is the stock price in the present, and S_{t-1} is the stock price one period back:

$$\text{Simple Return} = \frac{102 - 100}{100} = 2\%$$

That's a simple way of calculating the returns of a stock, but it contains a non-obvious problem, which is the asymmetry between positive and negative returns. Let's use a simple example to illustrate. Imagine that the stock price is \$100, and in the next day it goes to \$105, the simple return formula will indicate:

$$\text{Simple Return} = \frac{105 - 100}{100} = 5\%$$

However, if the stock price is at \$105, and it goes to \$100 in the next day, the formula will return:

$$\text{Simple Return} = \frac{100 - 105}{105} = -4.76\%$$

Here lies the problem of simple returns, which is that even though the stock price varies \$5 in both situations, the returns are not symmetrical because they are not computed on the same base. To solve this problem, we need to use the *continuously compounded returns* or the *log returns*. To understand the continuously compounded return, we need to review the basics of natural logarithms.

The logarithm is the inverse function of exponentiation. That means the logarithm of a given number N is the exponent to which another fixed number (the base a) must be raised to produce that number N:

$$\text{Log}_a N = x \qquad N = a^x$$

For example, the logarithm of 8 with base 2 is 3:

$$\text{Log}_2 8 = 3 \qquad 8 = 2^3$$

The *natural logarithm* is just a logarithm with base equal to *e*, which is a mathematical constant known as Euler's Number equal to 2.71828. The natural log is used to calculate the continuous return in finance using the following formula:

$$\text{Log Return} = \ln\left(\frac{S_t}{S_{t-1}}\right)$$

If we use this formula, when price goes from \$100 to \$105, we get:

$$\text{Log Return} = \ln\left(\frac{105}{100}\right) = \mathbf{4.879\%}$$

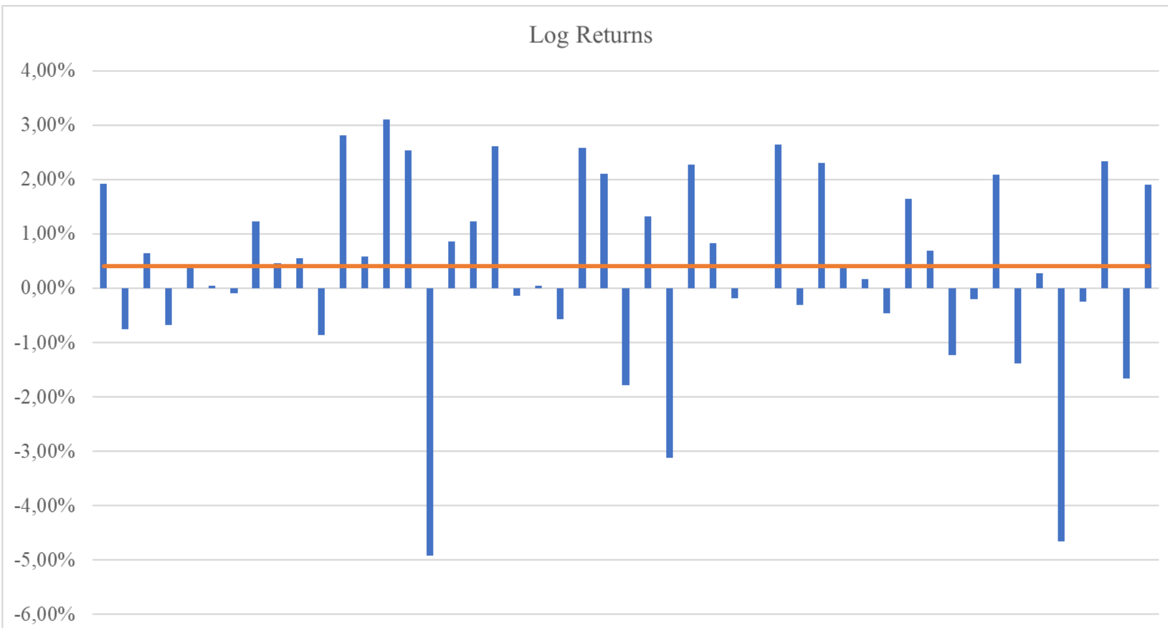
And when price goes from \$105 to \$100, we get:

$$\text{Log Return} = \ln\left(\frac{100}{105}\right) = \mathbf{-4.879\%}$$

Observe now that returns are symmetrical. That happens because the compounding formula and the formula for *e* are almost identical. Now we need to develop the intuition for standard deviation since the historical volatility is the standard deviation of log returns. Before understanding standard deviation, we must understand the concept of ***variance*** because *standard deviation is nothing more than the square root of variance*. We'll get why that's the case in a moment. The formula for variance is:

$$\sigma^2 = \frac{1}{m}\sum_{i=1}^{m}(x_i - \mu)^2$$

To unpack this formula, we'll consider a series of log returns from the Apple stock price in a period of 50 days:



The blue bars show the log returns from a period of 50 days, and the orange line shows the mean of these log returns. Notice that some of the returns are above the average, while others are below the average. This helps to understand one of the parts in the variance formula:

$$(x_i - \mu)^2$$

The term x_i in this case represents each log return, and the term μ represents the mean of log returns. This part of the formula is measuring *the distance between each return and the mean*. However, since some returns are below the average, when we compute the difference, a negative number will appear. To solve this problem, *we square the difference*. Any negative number squared will result in a positive number. This also explains why we have to take the square root of the result after we account for the other parts of the variance formula in order to find the standard deviation.

The summation notation simply means that we are going to sum all the squared differences starting from the first log return until m, which is the number of log returns that we have (49 in this case):

$$\sum_{i=1}^{m} (x_i - \mu)^2$$

The first term $\frac{1}{m}$ simply means that we are finding the average of the squared differences:

$$\sigma^2 = \frac{1}{m}\sum_{i=1}^{m} (x_i - \mu)^2$$

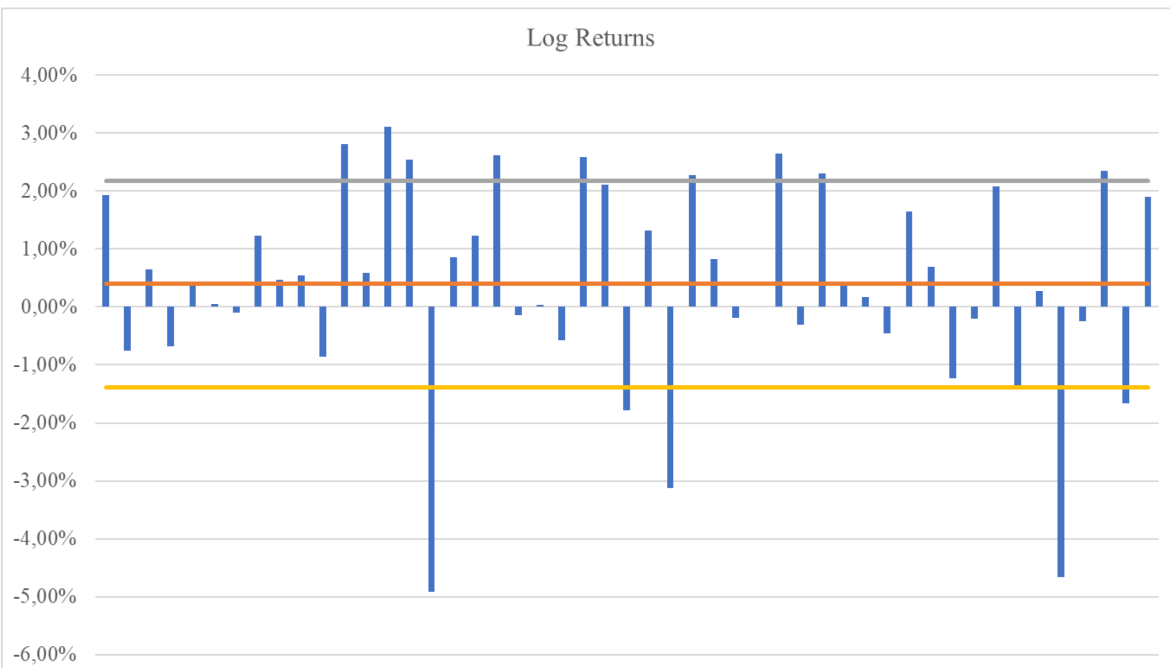
If we compute the variance for this example, we'll get:

$$\sigma^2 = 0.031692\%$$

Now all that we have to do to find the standard deviation is to take the square root of that value:

$$\sigma = 1.7802\%$$

If we plot the upper and lower standard deviations from the mean, we can have an idea about its meaning:



In other words, the standard deviation of log returns gives an idea about how much the individual data points stretch away from the mean of log returns. This example also serves to show that we can make a simplification in the formula for standard deviation of log returns. Notice that the mean is very close to zero. That's the case in most log returns from financial data. In other words, we don't need to measure the difference between each data point and the mean if the mean is zero. We just need to square the log returns:

$$\text{Variance} = \sigma^2 = \frac{1}{m}\sum_{i=1}^{m} x_i^2$$

$$\text{Standard Deviation} = \sigma = \sqrt{\frac{1}{m}\sum_{i=1}^{m} x_i^2}$$

This calculation we just did is called a Moving Average Model for volatility. It's the simplest way to calculate the historical volatility, and it's not used in practice because it has a few limitations. It's a moving average model because we are calculating the standard deviation of the past 49 log returns, and as a new daily price enters the model, the oldest log return leaves the model. The limitation of the Moving Average model for volatility is that *all log returns have the same weight*, which is not realistic. In practice, the log return from the day before is much more relevant than older log returns. The next step to improve the model is to assign different weights to the log returns in what's called the Exponentially Weighted Moving Average Model (EWMA).

Before we move on, let's make a change in the mathematical notation. So far, we have called the periodic log returns as x_i, but we'll change that to u_{n-1} to conform with the mathematical notation that is commonly used in finance. That's because we want to estimate today's variance as a function of yesterday's variance and yesterday's log return. The formula for variance in the moving average model then becomes:

$$\sigma_n^2 = \frac{1}{m}\sum_{i=1}^{m} u_{n-1}^2$$

In simpler terms, the moving average model for variance is the average of squared log returns. The volatility (standard deviation) is simply the square root of the variance. The next step is the EWMA model that assigns different weights to different log returns depending on how recent they are. The model is called exponentially weighted moving average because it assigns larger weights to more recent log returns, and these weights decay exponentially with older log returns. The way this is done is by the determination of a parameter called lambda (λ), which is going to determine the weight size of the most recent log return, and all the older weights as well.

In this example, we'll assume that lambda is 90%, so the weight for the squared log return from t-1 is going to be 1 - λ, and the previous squared log returns are going to be weighted by lambda as well. For example, the weight at t-2 is going to be the weight at t-1 times lambda:

Time	Stock Price	Log Returns	Squared Log Returns	Weights	Weighted Squared Log Returns
t-50	313,14	-	-		
t-49	319,23	1,93%	0,037100%	0,06%	0,000024%
t-48	316,85	-0,75%	0,005600%	0,07%	0,000004%
t-47	318,89	0,64%	0,004119%	0,08%	0,000003%
t-46	316,73	-0,68%	0,004619%	0,09%	0,000004%
t-45	318,11	0,43%	0,001890%	0,10%	0,000002%
t-44	318,25	0,04%	0,000019%	0,11%	0,000000%
t-43	317,94	-0,10%	0,000095%	0,12%	0,000000%
t-42	321,85	1,22%	0,014940%	0,13%	0,000020%
t-7	388,00	-1,39%	0,019315%	5,31%	0,001026%
t-6	389,09	0,28%	0,000787%	5,90%	0,000046%
t-5	371,38	-4,66%	0,217015%	6,56%	0,014238%
t-4	370,46	-0,25%	0,000615%	7,29%	0,000045%
t-3	379,24	2,34%	0,054867%	8,10%	0,004444%
t-2	373,01	-1,66%	0,027437%	9,00%	0,002469%
t-1	380,16	1,90%	0,036051%	10,00%	0,003605%
t					
			Lambda	90%	

Notice that we are considering a lambda of 90%, so the weight at t-1 (yesterday) is equal to 1- λ, which equals to 10%. The weight at t-2 is simply the weight at t-1 times lambda, which equals to 9%. We do this recursively until t-49. Observe that I collapsed the lines between t-7 and t-42 so it's easier to see here. More importantly, notice that the more recent returns have more weight than the older returns. Another important thing to notice here is that if we sum the weights, they will eventually sum to 100%. To calculate the estimate for today's variance at t, first we need to multiply each squared log return by its weight. The result can be seen in the column named Weighted Squared Log Returns.

In the simple moving average model, the variance is simply the average of squared log returns. In the EWMA model, since we are assigning different weights to the squared log returns, and the series of weights eventually sums up to 100%, we sum all the weighted squared log returns to arrive at the variance:

	Variance	Standard Deviation
MA Model	0,031692%	1,7802%
EWMA Model	0,036541%	1,9116%

Remember that volatility equals standard deviation, which is the square root of variance. Observe that the volatility in the EWMA model is higher than in the simple MA model. That's because the EWMA model is assigning heavier weights to more recent returns, and these returns were larger than normal. For example, at t-3 we have a log return of 2,34% and at t-5 we have a log return of -4,66%. This is a more intuitive way of calculating the estimate for today's variance in the EWMA model. We can also calculate it using the recursive formula as follows:

$$\sigma_n^2 = \lambda\sigma_{n-1}^2 + (1 - \lambda)u_{n-1}^2$$

In other words, the estimate for today's variance is a function of yesterday's variance in the EWMA model times lambda + yesterday's squared return times (1 – lambda):

Time	Stock Price	Log Returns	Squared Log Returns	Weights	Weighted Squared Log Returns	Weights - 1	Weighted Squared Log Returns - 1
t-50	313,14	-	-				
t-49	319,23	1,93%	0,037100%	0,06%	0,000024%	0,07%	0,000026%
t-48	316,85	-0,75%	0,005600%	0,07%	0,000004%	0,08%	0,000004%
t-47	318,89	0,64%	0,004119%	0,08%	0,000003%	0,09%	0,000004%
t-46	316,73	-0,68%	0,004619%	0,09%	0,000004%	0,10%	0,000004%
t-45	318,11	0,43%	0,001890%	0,10%	0,000002%	0,11%	0,000002%
t-44	318,25	0,04%	0,000019%	0,11%	0,000000%	0,12%	0,000000%
t-43	317,94	-0,10%	0,000095%	0,12%	0,000000%	0,13%	0,000000%
t-42	321,85	1,22%	0,014940%	0,13%	0,000020%	0,15%	0,000022%
t-7	388,00	-1,39%	0,019315%	5,31%	0,001026%	5,90%	0,001141%
t-6	389,09	0,28%	0,000787%	5,90%	0,000046%	6,56%	0,000052%
t-5	371,38	-4,66%	0,217015%	6,56%	0,014238%	7,29%	0,015820%
t-4	370,46	-0,25%	0,000615%	7,29%	0,000045%	8,10%	0,000050%
t-3	379,24	2,34%	0,054867%	8,10%	0,004444%	9,00%	0,004938%
t-2	373,01	-1,66%	0,027437%	9,00%	0,002469%	10,00%	0,002744%
t-1	380,16	1,90%	0,036051%	10,00%	0,003605%	Variance t-1	**0,036595%**
t				Variance	**0,036541%**		
			Lambda	90%			
				Variance	Standard Deviation		
			MA Model	0,031692%	1,7802%		
			EWMA Model	0,036541%	1,9116%		

To use the recursive formula, we need to know the estimate for variance at t-1, so we simply repeat the weighting process one period back. Plugging the recursive formula for the EWMA model, we get the same exact result from before:

$$\sigma_n^2 = \lambda\sigma_{n-1}^2 + (1 - \lambda)u_{n-1}^2$$

$$\sigma_n^2 = 90\%\ 0.036595\% + (1\text{-}90\%)0.036051\%$$

$$\sigma_n^2 = 0.036541\%$$

Understanding the recursive formula for the EWMA model is important because the GARCH(1,1) model, which is the formula used to forecast volatility, is just the EWMA model with an extra step. The GARCH(1,1) formula goes as follows:

$$\sigma_n^2 = \gamma V_L + \alpha u_{n-1}^2 + \beta\sigma_{n-1}^2$$

Let's unpack the formula to understand its similarity with the EWMA model. Observe that the estimate for today's variance is still a function of yesterday's variance (σ_{n-1}^2), and yesterday's squared log return (u_{n-1}^2). The extra step is the addition of a long-run variance V_L in the formula. Notice also that each term has a weight assigned to it. Before talking about how to calculate the long-run variance and the different weights (gamma, alpha, and beta), we must apprehend that the GARCH(1,1) attempts to model the idea of mean reversion. In other words, the volatility tends to revert back to a long-run average.

The way we calculate the long-run variance is simply by calculating the sample variance for the log returns in the series. The weights gamma, alpha, and beta are usually 5%, 5%, and 90% respectively (they must sum to 100%), so we simply plug the GARCH(1,1) formula for with these parameters:

Time	Stock Price	Log Returns	Squared Log Returns	Weights	Weighted Squared Log Returns	Weights - 1	Weighted Squared Log Returns - 1
t-50	313,14	-	-				
t-49	319,23	1,93%	0,037100%	0,06%	0,000024%	0,07%	0,000026%
t-48	316,85	-0,75%	0,005600%	0,07%	0,000004%	0,08%	0,000004%
t-47	318,89	0,64%	0,004119%	0,08%	0,000003%	0,09%	0,000004%
t-46	316,73	-0,68%	0,004619%	0,09%	0,000004%	0,10%	0,000004%
t-45	318,11	0,43%	0,001890%	0,10%	0,000002%	0,11%	0,000002%
t-44	318,25	0,04%	0,000019%	0,11%	0,000000%	0,12%	0,000000%
t-43	317,94	-0,10%	0,000095%	0,12%	0,000000%	0,13%	0,000000%
t-42	321,85	1,22%	0,014940%	0,13%	0,000020%	0,15%	0,000022%
t-7	388,00	-1,39%	0,019315%	5,31%	0,001026%	5,90%	0,001141%
t-6	389,09	0,28%	0,000787%	5,90%	0,000046%	6,56%	0,000052%
t-5	371,38	-4,66%	0,217015%	6,56%	0,014238%	7,29%	0,015820%
t-4	370,46	-0,25%	0,000615%	7,29%	0,000045%	8,10%	0,000050%
t-3	379,24	2,34%	0,054867%	8,10%	0,004444%	9,00%	0,004938%
t-2	373,01	-1,66%	0,027437%	9,00%	0,002469%	10,00%	0,002744%
t-1	380,16	1,90%	0,036051%	10,00%	0,003605%	Variance t-1	**0,036595%**
t				Variance	**0,036541%**		
			Lambda	90%			
				Variance	Standard Deviation		
			MA Model	0,031692%	1,7802%		
			EWMA Model	0,036541%	1,9116%		
			GARCH(1,1)	0,0362757%	1,9046%		
			Gamma	5%			
			Alpha	5%			
			Beta	90%			
			Long-Run Variance	0,030753%			

$$\sigma_n^2 = \gamma V_L + \alpha u_{n-1}^2 + \beta \sigma_{n-1}^2$$

$$\sigma_n^2 = 5\% \; 0.030753\% + 5\% \; 0.036051\% + 90\% \; 0.036595\%$$

$$\sigma_n^2 = 0.0362757\%$$

The estimate for today's volatility is simply the square root of the estimate for today's variance:

$$\sigma_n = 1.9046\%$$

- How Institutional Traders Forecast Volatility

The way institutional traders predict volatility is surprisingly simple. The forecast is made by comparing the *annualized* volatility predicted by the GARCH(1,1) model and the implied volatility of the option being traded. Before understanding the comparison, let's learn how to annualize the volatility given by the GARCH(1,1), which is daily volatility. All we need to do is to multiply the daily volatility by the square root of 252, which is the number of business days in a year. We have to be careful here because depending on the country, the annualized volatility is measured with calendar days. The annualized volatility given by the GARCH(1,1) is:

$$1.9046\% \sqrt{252}$$

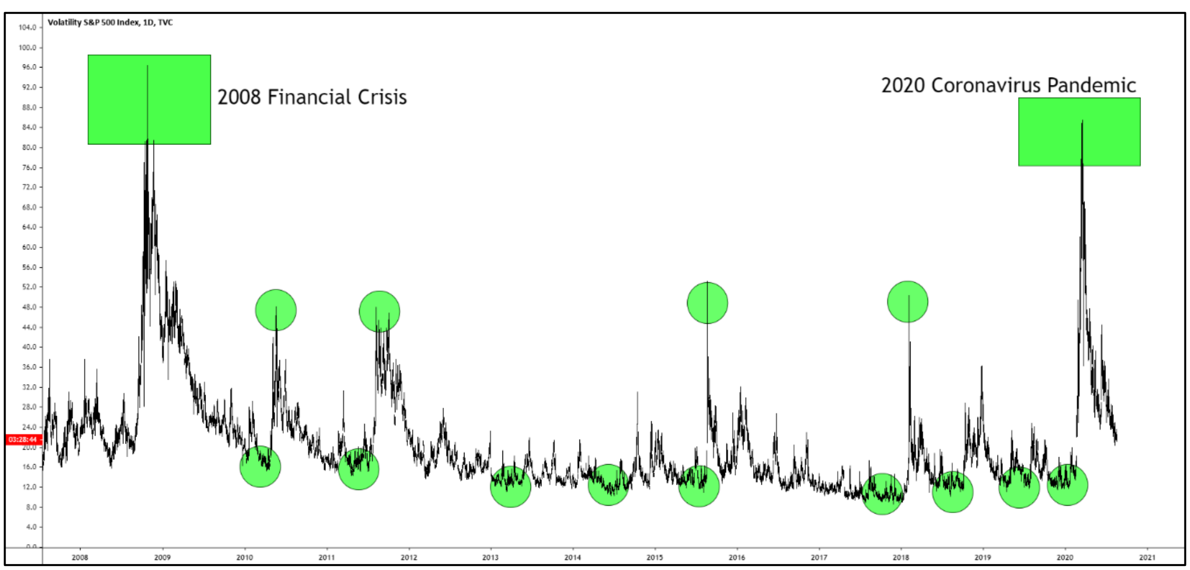
$$30.23\%$$

This is a necessary step because the implied volatility of options is also measured on an annualized basis. ***<u>When the implied volatility of an option is lower than the volatility given by the GARCH(1,1) model, an institutional trader assumes that volatility is cheap</u>***. In other words, the GARCH(1,1) is predicting that the volatility will rise. That's the green light for a delta hedge operation. Needless to say, these methods are just estimations. They are *not* a guarantee of profit, but they have very important roots. We are comparing different ways of measuring volatility from two different Nobel Prizes in Economics. The GARCH is a generalization of a model called ARCH (autoregressive conditional heteroskedasticity), and the implied volatility is extracted from the Black-Scholes model. Both the ARCH model and the Black-Scholes model won the Nobel Prize in Economics.

One note about econometric models is that they work when there is a certain normality in the market. In extreme events like the 2008 financial crisis and in the more recent 2020 coronavirus pandemic that created too much volatility in the markets, such econometric models like the GARCH(1,1) will get distorted and give false signals. The best thing to do in such scenarios is to wait for the markets to normalize.

- The VIX

The VIX is the *volatility index* of the S&P500. It's the average of implied volatilities of the options in the S&P500 index. By looking at the daily VIX chart, we can spot a few interesting properties that we don't usually notice on the price charts of other instruments like stocks, currencies and so on:



Notice how volatility is *stationary* over the years, meaning that it's a constantly sideways market. A stationary market has one property that makes it more predictable than trending markets. Every time such a market reaches its lower range, we can expect it to rise. Notice that every time the VIX hits the baseline 8% to 12% volatility, it started to rise immediately after. In some cases, there are larger peaks like in the 2008 financial crisis where the VIX ended up hitting 96%, and in the 2020 coronavirus pandemic where the VIX hit about 85%.

Notice that before the coronavirus pandemic, which caused an enormous amount of volatility in the markets, the VIX was near its low. That doesn't mean that the VIX predicted the pandemic. It simply means that the VIX is stationary, so we can expect the volatility to rise when it hits relative lows, and to fall when it hits relative highs. Notice also that between 2008 and 2020, there were many smaller peaks of volatility around the same 48% level, and in between those, many smaller peaks around 25%.

That of course represents an interesting idea to delta hedgers because the primary requirement of a delta neutral trade is to buy volatility when it's cheap. Every time the market returns to a certain relative volatility low, the trader can expect volatility to rise again. If we look closer to the chart, we can see that this happens quite frequently also. Here it's important to recall once again that a delta hedger *never* sells volatility because that implies a negative gamma position. A delta hedger only buys volatility.

It's important to point out also that looking at the relative highs and lows of the VIX is just an anecdotal way of forecasting volatility, and it's *not* the way institutional traders approach the problem. An institutional trader will still look at the comparison between the GARCH(1,1) model in relation to the implied volatility of the option being traded because that's a more sophisticated and robust way of forecasting what's going to happen with volatility. With that said, we can indeed see that the market *sometimes* respects the relative volatility highs and lows:

Vega Scalping

Beyond Gamma Scalping, there is also the possibility of performing Vega Scalping, which is exactly what it sounds like. Progressively taking advantage of small increases in the implied volatility of an option. Vega scalping is quite similar to the simple Volatility Trade where only the portfolio Delta is neutralized. The difference dwells in the time horizon that the trade is made, and in the entry and exit points of the operation. Before diving into the specific aspects of this trade, we must pay attention to certain properties of the implied volatility of an option to understand why Vega Scalping makes sense.

- Nonstationary and Stationary Implied Volatility

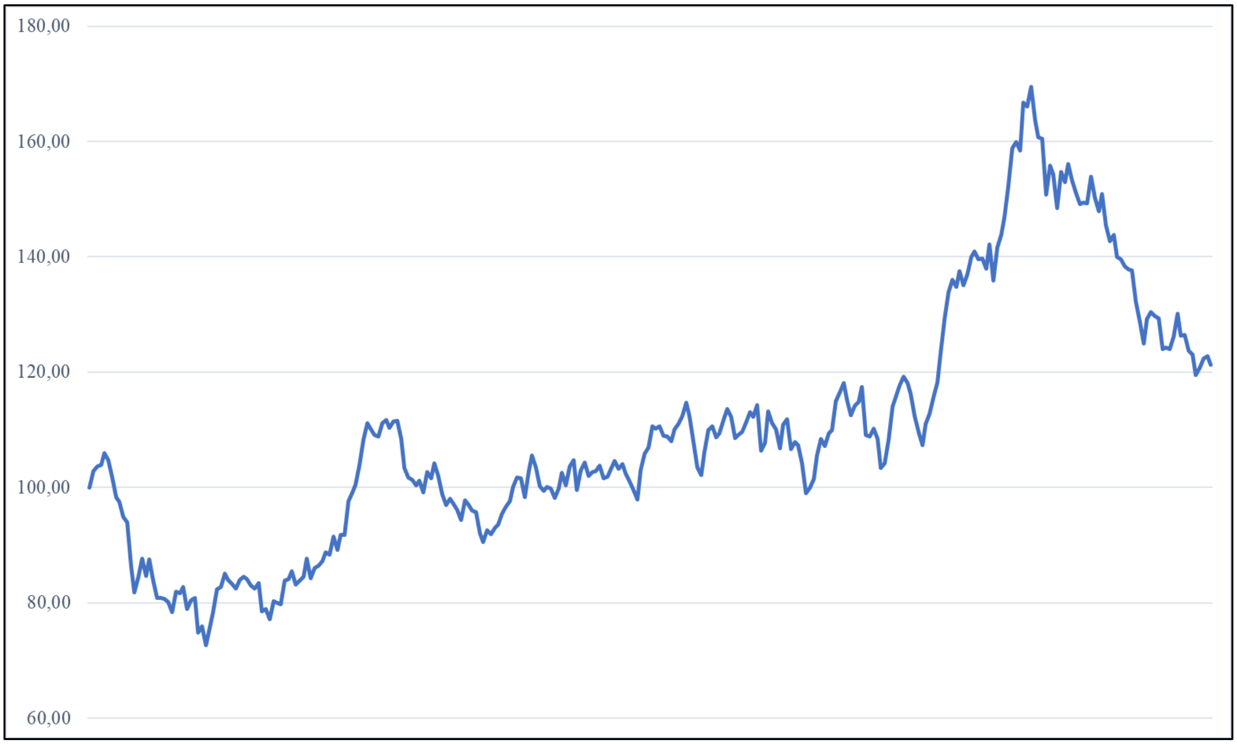
In the simple volatility trade, you learned that when the implied volatility of the option being traded is below the value predicted by the GARCH(1,1) model, it means that the implied volatility is cheap, and it will probably rise in the next days. The thing is that implied volatility tends to rise from one day to the other, *but throughout the day, it tends to be stationary most of the times*. Here we have to learn an important concept in econometrics called ***stationarity***.

In technical terms, a stationary time series has constant mean and constant standard deviation, meaning that it always oscillates around a fixed axis. In practical terms, a stationary time series is like a sideways market. It has no trend. The interesting property about stationary timeseries is that *they are predictable*. That happens because if the series oscillates around a fixed mean, we can predict when it will come back to this mean when it escapes a certain number of standard deviations from the mean. Technical traders mistakenly attempt to apply this idea in nonstationary markets, which have variable mean. That's like trying to score a goal with moving goalposts.

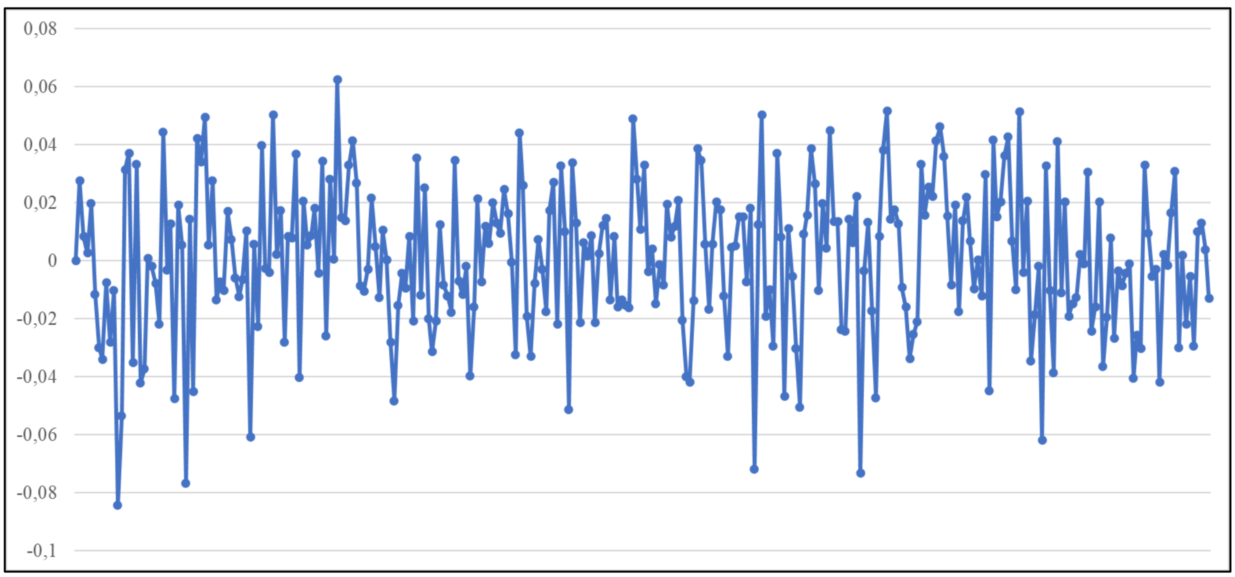
The implied volatility of options tends to be stationary throughout the day, which means that when the intraday implied volatility stretches away a certain number of standard deviations from its mean, it will tend to revert back to this mean. That means we can scalp the Vega of an option in the intraday time horizon. There are a few peculiarities about the Vega scalping approach. The first is that we never short sell volatility, so the Vega scalping should only be done when the implied volatility is two standard deviations below the intraday mean.

An interesting aspect is that since the operation is done D+0 (intraday), there is no Theta risk to worry about. Beyond the Vega, it's likely that gains in Gamma will also occur, even though the focus is on the Vega. Since the operation is Delta neutral, it doesn't matter where price is going, we only care about the implied volatility in here. Since we are long on volatility, *any* events that shock the market unexpectedly will go in our favor. Before moving one to the practical and operational aspects of Vega Scalping, let's apprehend the difference between stationary and nonstationary timeseries graphically.

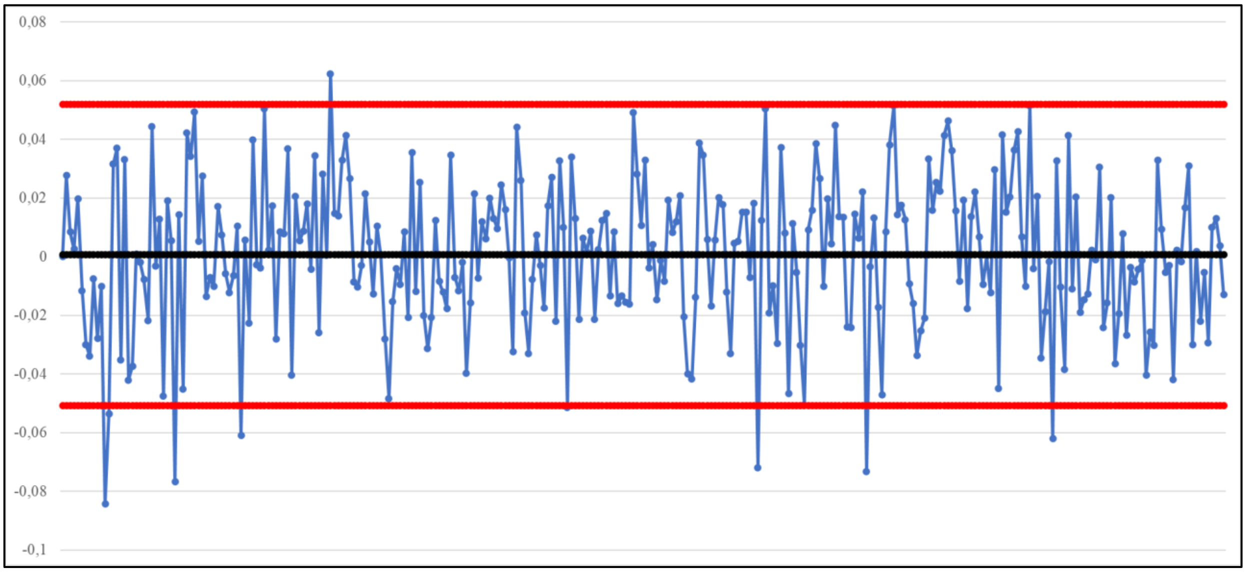
In the graph below, we have an example of computer generated timeseries that follows a Brownian motion just like price does. Just by looking at the chart, we can see that there is some unpredictability about how the highs and lows vary, and we can notice how the series doesn't have a fixed mean (if we plot a moving average in there, we'll see it going up and down). This is an example of nonstationary timeseries, which cannot be predicted using the mean reversion approach.



In the graph below however, we have a plot of the log returns from the timeseries above:



There is an explicit difference between the two graphs. Notice how the log returns fluctuate around a fixed mean, which is very close to zero in this case. This is an example of timeseries that can be seen under the light of mean reversion because if the series oscillates around a fixed mean, every time it escapes a certain number of standard deviations from this mean, we can expect it to return to the mean without that problem of "moving goalposts" that occur in nonstationary timeseries. Below you can see the log return series with its mean and two standard deviations from the mean plotted:



When we plot the two standard deviations from the mean both for the upside and for the downside, it becomes clear why a stationary timeseries can be seen under the paradigm of mean reversion. Just as a parenthesis, we use *two* standard deviations from the mean because of the empirical rule in statistics, which claims that around 68% of data points will fall within one standard deviation from the mean, 95% of data points will fall within two standard deviations from the mean, and 99.7% of data points will fall under three deviations from the mean. We can see that this is in fact the case in this series. We have 300 data points, and 292 of them are captured by the space between the mean and the two standard deviations.

The bottom line here is simple, every time the series goes outside the two standard deviation barriers, we can expect it to return to the mean, but remember, *this sort of thing only works if the series is stationary*. It completely fails in nonstationary markets. That's a mistake that many retail traders commit when trying to trade the Bollinger Bands indicator on price charts. Such traders are not aware that the mean reversion strategy only works in stationary markets. If you pay attention, the black and red lines we plotted on the log return series are exactly like the Bollinger Bands, but they have fixed mean and fixed standard deviation.

All of this serves to illustrate the fact that, in most days, the implied volatility of options *tends* to be stationary throughout the day. Not always of course. This is the origin of the Vega Scalping strategy. We can count on the fact that the implied volatility of an option will remain stationary throughout the day, so every time the implied volatility escapes two standard deviations below the mean, we can enter a Vega Scalping position based on the expectation that the implied volatility will return to its mean.

However, due to the particular way derivatives work, there are some peculiarities to this type of operation.

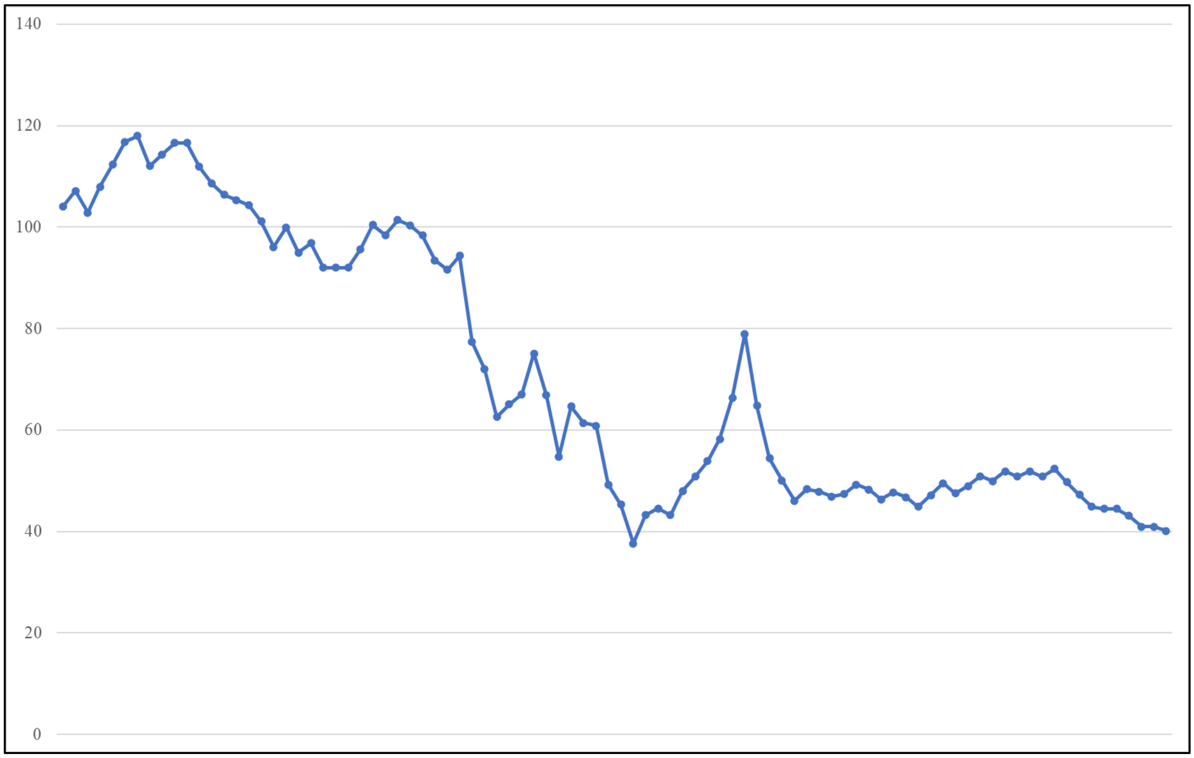
Vega Scalping Rules

As it was already mentioned before, Vega Scalping is done D+0 (intraday), so we don't have to worry about the Theta risk. That leaves us with Delta, Gamma, and Vega. If we buy options, we will enter a positive Gamma position, so any movement away from the hedge point will positively affect the P/L, and if price stays still, nothing will occur to the P/L from the perspective of Gamma. Delta is easy to deal with. In other words, in the Vega Scalping paradigm, we only worry about the Vega, and depending on what happens, we will get the added benefit from gains in Gamma as well, but not always.

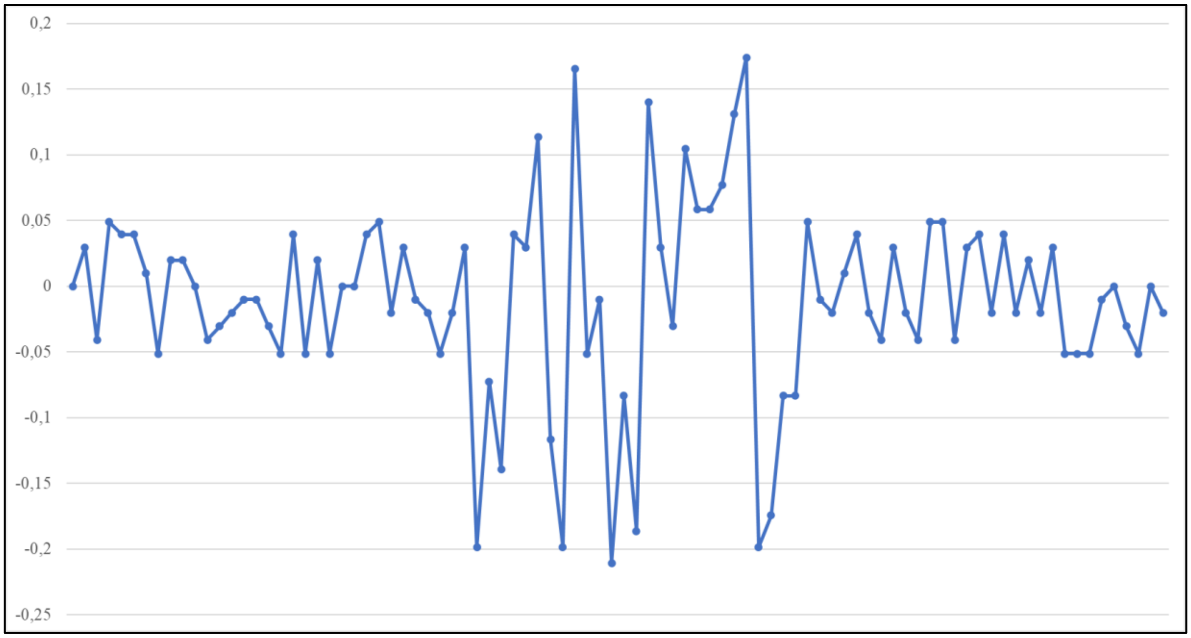
The peculiar thing about Vega Scalping is that we can only buy volatility, so we only enter the Vega Scalping operation when the intraday implied volatility goes two standard deviations *below* its mean. If we try to scalp the Vega when it goes two standard deviations above its mean, we are basically short-selling volatility, which implies a negative Gamma position, and you already know that is a terrible idea. Notice that derivatives are completely different than the spot market. Buying and selling in the derivatives market is asymmetrical.

Another point to consider here is the relationship between Greeks and time. We know that the longer the expiration, the more Vega an option will have, and the less Gamma and Theta it will have. However, the Vega Scalping operation is relatively quick, which implies that the option being traded needs tons of liquidity to work, and sometimes options with longer expiration lack this liquidity. Perhaps the most problematic thing about the Vega Scalping operation is the degree of stationarity of the implied volatility.

Even though the intraday implied volatility does tend to be stationary, sometimes it oscillates around a fixed mean, but it has variable standard deviation, which will create the problem of moving goalposts in a similar way that occurs in nonstationary timeseries, but in a less dramatic degree let's say. To understand this, consider the following series:



Now let's plot the log return of this series, and a pattern will become obvious:



Notice that for the first and third thirds of the series, we have a relatively small standard deviation. In the second third, the standard deviation increases dramatically. Even though the series fluctuates around a fixed mean, the variance, and therefore the standard deviation is variable. This can create problems in the Vega Scalping operation obviously. Nonetheless, this is a better problem than dealing with mean reversion in nonstationary timeseries where not only the standard deviation is variable, but the whole series fluctuates around a variable mean.

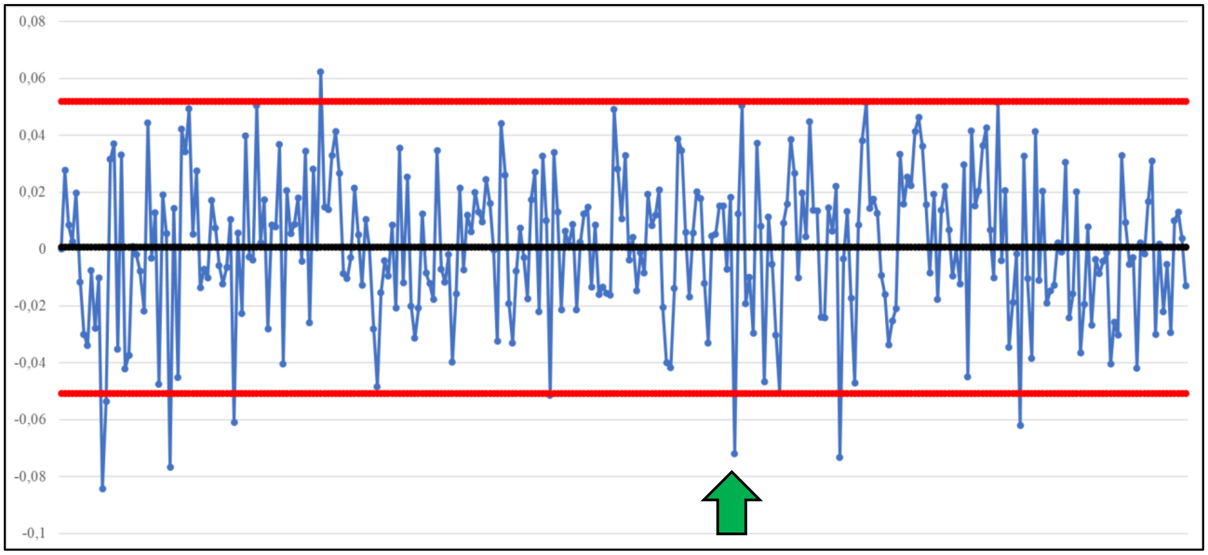
The initial rules for Vega Scalping go as follows:

1) Choose an ATM option.
2) Find the sweet spot between longer expiration and liquidity.
3) Observe the intraday implied volatility chart or table.
4) Buy the implied volatility via Delta Hedge when it's two standard deviations below its mean.
5) Never short sell volatility when it's two standard deviations above its mean.
6) Exit when the implied volatility returns to the mean.
7) The stop loss can be the three or four standard deviations below the mean. It will depend on your risk aversion.

In the other books, I mentioned the fact that Delta Hedging is a prerequisite for volatility strategies. It's not a strategy per se. The Vega Scalping operation is not an exception. We buy volatility by Delta Hedging because we want to eliminate the directional risk of the market, and only focus on the Vega risk. By the way, the risk in an operation like this is the market being quiet for too long after you entered the trade, which will make the implied volatility fall too much and hit your stop loss.

Vega Scalping in Practice

Let's imagine, for simplicity sake, that the chart below is the intraday implied volatility plot of an ATM option that is two months from expiration:



Let's also say that when you see the implied volatility go below the two standard deviations barrier as you can see in the green arrow, you decide to enter a Vega Scalping operation. The next step is quite simple. We can buy volatility via Delta Hedge by either neutralizing the Delta using the underlying stock or by buying a specific combination of call and put options. In this example, let's use a combination of call and put options:

Option Strategy Simulator — macroption

-1w	-1d	19/10/2020	+1d	+1w	-5%	-1%	Und Price	+1%	+5%	Div Yield	2,00%	Show Values & Greeks	-5%	-1%	Volatility	+1%	+5%
Now		11:33:25	Expiration		-1	-0.1	50,00	+0.1	+1	Int Rate	1,75%	Total	-1	-0.1	20,00%	+0.1	+1

Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price	Value	P/L	Delta	Gamma	Theta	Vega	Rho
1	50	Dec 2020	60,20	50	Call	1,60387	-8.019	1,60387	8.019	0	2.562	489	-66	403	198
2	52	Dec 2020	60,20	50	Put	1,62441	-8.513	1,62441	8.513	-0	-2.538	513	-71	423	-223
3		Jun 2020			None										
4		Jun 2020			None										
5		Jun 2020			None										
							-16.533		16.533	0	25	1.002	-136	826	-25

We enter a Delta neutral and positive Gamma position with the expectation that the implied volatility will return to its mean, which it does if we observe the graph. Let's assume that the distance between the two standard deviations below the mean and the mean represent 2% in implied volatility. When the implied volatility does return to the mean, we get:

Option Strategy Simulator — macroption

-1w	-1d	19/10/2020	+1d	+1w	-5%	-1%	Und Price	+1%	+5%	Div Yield	2,00%	Show Values & Greeks	-5%	-1%	Volatility	+1%	+5%
Now		11:33:25	Expiration		-1	-0.1	50,00	+0.1	+1	Int Rate	1,75%	Total	-1	-0.1	22,00%	+0.1	+1

Leg	Position	Expiration	Days	Strike	Type	Initial Price	Initial CF	Price	Value	P/L	Delta	Gamma	Theta	Vega	Rho
1	50	Dec 2020	60,20	50	Call	1,60387	-8.019	1,76518	8.826	807	2.571	445	-72	403	197
2	52	Dec 2020	60,20	50	Put	1,62441	-8.513	1,78572	9.359	845	-2.528	466	-78	423	-224
3		Jun 2020			None										
4		Jun 2020			None										
5		Jun 2020			None										
							-16.533		18.185	1.652	43	911	-150	826	-26

The position now has a $1,652 profit due to the increase in implied volatility. Here I'm not accounting for possible gains in Gamma that would probably increase the P/L too. This is a very good profit considering that the entire position required $16,533 to be opened. Obviously, the higher the profit, the higher the risk. The larger the standard deviation, the higher the risk also. Naturally, performing such operation using only derivatives is a leveraged way of doing it, and therefore, a lot riskier. You can do the same operation with less risk by neutralizing the Delta using the underlying stock.

There are days when such opportunities will appear multiple times, and there are days where it won't appear at all or the implied volatility won't be stationary. The important thing about Vega Scalping is that is based on the fact that, in the majority of days, the implied volatility tends to be stationary, which is equivalent to saying that, in most days, the implied volatility is a sideways market, and therefore, easily predictable through mean reversion. Another interesting thing about Vega Scalping is that it is an antifragile strategy too. If you enter a Vega Scalping trade, and while you are in the middle of the trade some major news comes out to agitate the market, *you will make money without a shadow of doubt because the implied volatility will certainly spike up with any piece of breaking news*. Since you are long on volatility, the positive Vega will make you money. This is also why it's never a good idea to short-sell volatility.

The only way to lose money with Vega Scalping is if the market doesn't move for too long, which will make the implied volatility fall too much to hit your stop loss, but we can certainly rely on the fact that the markets don't like to stand still for too long. Notice that Delta Hedging is common among all volatility strategies. What changes is the time horizon, the entry, and the exit of the strategies.

- Important Practical Aspects of Volatility Trading

1) Underlying Stock Price & Implied Volatility Alters the Required Margin

As we saw in one of the examples above, the underlying stock price changes the requires margin for a trade assuming you are going to neutralize the portfolio Delta using the stock instead of the doing it only with options. If the underlying stock price is $30 for example, it will require much less margin than it the stock price was $300. A similar thing happens with implied volatility. The higher the implied volatility, the higher the option premium, and therefore, the more margin is required to create a Delta hedge.

2) It Can Be Difficult to Neutralize the Portfolio Delta with Small Option Contracts

The larger the number of option contracts in the portfolio, the easier it is to fine tune the portfolio Delta, so if you are dealing with small contracts, it can be difficult to neutralize the portfolio Delta, and that will create a position that is either slightly long or slightly short and/or slightly exposed to changes in implied volatility. It's not ideal, but you have to keep a close eye on this type of thing.

3) You Need an Algorithm to Execute These Trades

Most professional options trading platforms offer some sort of algorithm to execute trades automatically. In the case of simple volatility trading, this is a requirement because we have to execute two orders, and they must be executed simultaneously. Beyond that, depending on the size of your trades, the algorithm will also slice the operation in several chunks to decrease the risk of liquidity or trapping. The bottom line is that in each price, there is a specific equilibrium that neutralizes the portfolio Delta, and if you attempt to do all of this manually, there is a good chance you'll mess it up.

Just imagine what happens if the market is agitated, and you try to create a Delta/Vega hedge. You buy a certain number of call options, and by the time you take to create the other side of the hedge by either short selling the stock or buying put options, the market has already moved significantly. An algorithm will make calculations and execute the orders much faster and much more reliably. The algorithm is just an execution algorithm though. It's not going to analyze the market for you.

In the case of a Vega Scalping trade, the necessity for an algorithm is also very high because the implied volatility of options can change in a matter of seconds, so there is no way that a human can make the calculations fast enough to enter and exit the trade. Beyond that, opening and closing the Vega Scalping trade, which involves two simultaneous trades can only be done with an execution algorithm.

4) Gamma, Theta, Vega & Time

Gamma and Theta are inversely proportional to the time until the expiration of an option, which means that the closer the option is to the expiration date, the more pronounced the effect of these two Greeks. Vega is directly proportional to the time until expiration, meaning that the closer an option is to its expiration date, the less pronounced the effect of Vega. If you want to perhaps take more advantage of the Gamma effect, and be less exposed to the Vega, you want to choose an option that is closer to expiration. Don't forget that you'll have to deal with a more powerful Theta also.

5) Be Careful with Options with Less than 10 days until Expiration

With less than 10 days until expiration, the Black-Scholes model can produce certain distortions that can be dangerous. Gamma and Theta will become extremely pronounced for example. That obviously can work for or against you, but the overall risk will increase.

6) Avoid Low Liquidity Options at All Costs

Liquidity is a paramount aspect of volatility trading. Without enough liquidity, you can be stuck in positions where the Theta will keep eroding your profits or you'll have to pay an enormous spread to get out. Choose only the high liquidity options that will naturally have a small spread. In the case of Gamma Scalping, where the trader has to deal with options with longer expirations, the problem of liquidity can become more pronounced.

7) Practicing with the Excel Spreadsheet

Every operation must be carefully and surgically studied before it's put into practice. The best way of practicing in a first moment is to use an Excel spreadsheet like I used in the book. One important thing in this practice is learning how to use the Goal Seek function in Excel. Under the tab "Data", you'll find an option called "Hypothesis Testing", which will lead you to the “Goal Seek” function. With this function you’ll be able to quickly know how many contracts or shares are necessary to neutralize the Delta. There are three inputs in the Goal Seek function: The cell you want to change; the value you want it to change to, and what cell needs to change in order for the new value to be achieved.

Conclusion

Institutional trading methods work in a different level of sophistication when compared to the more common retail trading approaches of price action reading for example, and they offer a number of different challenges because of that complexity. Institutional trading methods are easier to rely on because their backbone is formed by a few Nobel Prize winning ideas, so in scientific terms, it's the best that the modern theory of finance has to offer. However, as with many other meaningful endeavors in life, these methods require constant study and improvement because there are many nuances to them. In the case of derivatives, things are far from being intuitive, so there are many situations and scenarios that must be studied in advance, and that are difficult to anticipate. In the case of statistical arbitrage using the cointegration model, it should be clear that the problem is in the study of economics, which is extensive and complex.

Interestingly enough, people who gravitate towards these methods are after a more objective way of trading, and that's precisely what they get, but that doesn't necessarily mean it's going to be easy. The good news about a good portion of these methods is that they have a lot less subjectivity than the price action reading methods, so even though the mathematics is heavier, decisions are more easily made, and it tends to be easier to control the psychological side of trading. Institutional trading methods remove a lot of the psychological pressure from trading because of the introduction of robust mathematics and statistics. You can see it as a tradeoff in this case.

The main goal of this series of books is to introduce retail traders to the more serious side of finance and econometrics, and to the Nobel Prize winning ideas that changed the financial markets in the last fifty years or so. You can be certain that the ideas outlined in these books represent the sweet spot between scientific breakthrough and retail trading applicability. Needless to say, the econometrics of financial time series goes way beyond the ideas presented in here, but these more complicated methods and models are outside the realm of applicability of the retail trader. They are mostly used in large funds because of their superior infrastructure and access to data.

In summary, this series of books is dedicated to show you the best paths that the modern theory of finance and econometrics has to offer to the retail trader, and put you on the starting position to learn the correct things about the financial markets. You don't have to trust random internet gurus that claim you can become an advanced trader in two lessons or pseudoscience once you understand where you need to go to find the correct information. Science is here to aid in this endeavor. The answers lie in mathematical finance, economics, and the econometrics of financial timeseries, among other fields of study that intersect these areas in many ways.